Detailed Description of Morphological Neuron Types

MCs

104 MCs were reconstructed (54 FR, 50 CR), these divided into three morphological subclasses and can be viewed here.

Morphology

MC somata were located in DL and most superficial PL surrounding the glomerular neuropil from all sides. Some cell bodies were located at considerable distance to the associated glomerulus. Somata shape varied: ovoid, triangular, globular and soma of very irregular outline were observed. The somata size varied across subclasses with cell size. The nuclei were large, predominantly ovoid with a frequently indented surface and contained several large nucleoli. The amount of cytoplasm immediately surrounding the nucleus differed substantially between cells and loosely related to the overall soma size. In MCs in which the nucleus was separated from the plasma membrane by only a thin layer, one or more large bulges of cytoplasm formed at either pole of the soma where a primary dendrite emanated. The perikaryon was densely populated with organelles: A myriad of ER cisternae, plenty of mitochondria, as well as lysosomes and endosomes at different stages were observed. The Golgi apparatus usually accumulated at the base of a primary dendrite. In larger cells with massive primary dendrites, organelles of the perikaryon spread far into the dendrite. A single primary cilium of medium length (0.5 – 4.3 µm, $\bar{x} \pm \sigma$: 2.7 ± 0.8 µm, n=46) arose from the soma or base of the main dendrite and projected into the surrounding extracellular space with no discernible, common direction among MCs.

Axons were identified in 70 MCs as neurites of relatively small size (0.3 – 2.0 µm, $\bar{x} \pm \sigma$: 0.8 ± 0.3 µm, n=55) and the lack of cellular organelles reaching into the emerging branch. In the remaining MCs, the soma was cut, or no axon could be unambiguously distinguished from dendrites. Especially among the centrally located MCs with well-preserved dendrite, many axons were only represented with few tenths of microns in the volume. Axons arose either from the soma (46/70) or one of the larger dendrite (24/70). Those axons emerging from a dendrite usually branched off from the proximal portion before the process ramified into a bushy dendrite. However, in rare cases a process also gave rise to several tufty side branches before tapering off and adopting an axonal appearance. Axon myelination was observed in 20 MCs, myelination started only at a length of several tenths of microns (> 30 microns). Most axons were projected laterally. No pronounced tendency of axons to align or form bundles was observed, although a few myelinated axons travelled together in small groups of fewer than five. Due to the limited size of the stack, it could not be determined whether the tendency to fasciculate was more pronounced as MC axons approached their exit points of the OB. Except for one large MC, local collaterals were only observed in members of the small MC subclass.

100 of 104 MCs and MC dendrites were strictly monoglomerular. One deviation from that rule was observed in MCs of glomerulus GL4. Here three MCs extended portions of their dendrite into the parts of the GL7 volume, which was less densely populated. In glomeruli with multiple compartments only large and intermediate MCs were found to cross compartment boundaries. However, there were six MCs that projected a neurite of large diameter outside of the imaged volume. Whether these MC innervated remote glomeruli or whether these dendrites reentered the volume to project into the same glomerulus as the rest of the cell's dendrite could not be determined. MC somata gave rise to

multiple dendritic branches (1 – 5, $\bar{x}$ ± σ: 3.3 ± 1.9). These branched asymmetrically and formed a tufted or bushy dendritic tree. The most distinct feature of MC dendrites was the high variance in diameter due to frequent dilations and shrinking of the branches along the entire dendritic tree. This was most pronounced in distal processes of higher branch order, where thin, sometimes filamentous dendrites abruptly broadened into large varicosities (Fig. 2A). The varicosities were irregular in shape, uneven or ragged in surface and occasionally bore various excrescences. Usually, the varicosities gave rise to several subbranches. The cytosolic composition distinguished these dendritic varicosities as sites of synaptic interaction (Fig. 2A). They contained at least one, but usually many dark mitochondria immersed in a large volume of cytosol. Further, an abundances of membrane structures of varying sizes and shapes was observed. Extensive membrane sacs with large lumen, often darkly stained, were identified as cisternae of the ER. Smaller vesicular structures, clearly larger than synaptic vesicles, some globular, some rather ellipsoid, were scattered through the cytosol, often intermingling with synaptic vesicle clouds. These could be clathrin-coated reuptake vesicles, endosomes, or secretory vesicles. Synaptic vesicles were globular and appeared larger than those found in neighboring IN. Within the varicosities, vesicles were so numerous that it was often difficult to distinctly outline a vesicle cloud and assign it to a specific synapse.

MC dendrites did not taper towards the tip branches, but rather terminated in rod-shaped, sometimes bulbous structures. These rods were usually the location of sensory input (Fig. 2A). Despite the absence of any presynaptic structure many vesicles of the size and shape of synaptic vesicles were loosely distributed in the cytosol of these rods. For a small subset of MCs in this dataset broad synapse types, i.e., inbound, outbound, reciprocal, and sensory synapses, and their respective distribution had been described previously(Masudi, 2016), confirming sensory input to target exclusively the distal most parts of MC dendrites, but revealing no pattern in the distribution of other synapse types. However, the identity of the synaptically connected neurons was not determined. In a qualitative examination MCs were found to connect to ten of 11 morphological classes of INs (Fig. 20A). Most of these connections were made through reciprocal, dendro-dendritic synapses MCs and INs.

Two classes of outbound synapses were apparent: synapses with small active zones and vesicle clouds were predominantly found in the large primary dendrite shafts and very voluminous varicosities. Larger synapses with large vesicle clouds were found primarily in the more distal varicosities.

Large MCs

This subclass comprised 19 FR and 18 CR. Large MCs were massive compared to other neurons in zebrafish OB (soma size: ø: 10 - 24 µm, $\bar{x}$ ± σ: 14.5 ± 2.8 µm). Characteristic for this subclass was a rough, uneven, and craggy surface of the entire cell except for the axon (ø: 0.75 - 2 µm, $\bar{x}$ ± σ: 1.1 ± .4 µm,). The surface often exhibited irregular indentations and fissures and bore pointy or frazzled excrescences, which did not resemble spines. Somata were mainly ovoid or triangular and smoothly transitioned into large, primary dendrites, which had a diameter of up to 8 µm at their base (ø: 1 – 8 µm, $\bar{x}$ ± σ: 3.3 ± 1.9 µm). In some cells additional smaller dendrites arose from the soma to form a single, tufty branch. The primary processes usually maintained a large diameter along their entire length and did not taper off quickly (Fig. 2A). The outline of the dendritic tree was roughly globular or bushy. Due to the heavy branching and the large varicosities, the morphology of the dendrites was highly complex and dense within the occupied volume.

Intermediate MCs

The intermediate MCs comprised 36 cells (17 FR, 19 CR). Their somata were of medium size (ø: 10 - 18 µm, $\bar{x} \pm \sigma$: 12.3 ± 1.6 µm) of irregular shape and usually small perikarya dominating. These MCs were characterized by an overall finer, more delicate appearance of the dendrite, which often occupied a smaller volume than that of large MCs. The processes were also uneven and varicose, but apart from occasional pointy excrescences, the surface of soma and dendrites appeared smooth. Intermediate MCs had two or more, long primary dendrites of medium diameter that started ramifying after a considerable path length. In a subset of intermediate MCs the dendrites bent back, resulting in an overall tufty and bushy dendrite, which was still large, but the volume was not densely occupied with subbranches. In other cells, they stretched out and diverged into a paint brush tip-like structure with the soma quite remote from the dendritic tuft.

Small MCs

Small MCs comprised 31 cells (21 FR, 10 CR fig. 14B). The somata of these cells were round or triangular and small and volume (ø: 8 - 14 µm, $\bar{x} \pm \sigma$: 10.6 ± 1.6 µm). One noteworthy feature of small MCs was the morphology of the axon. Axons emerged as fine processes from the soma or, rarely, the primary dendrite. Local collaterals, which remained in the GL but occupied neuropil volumes distinct from the glomerulus innervated by the dendrite were observed in ten small MCs. These axons bore bulbar varicosities, often with short, pointy, filamentous processes. These varicosities were quite distinct from dendritic varicosities and often resembled the varicose enlargements observed at the axons of the GLIN1s (Fig. 4B). Both pre- and post-synaptic sites were present in these varicosities. No myelination was detected in small MC axons. Small MCs gave rise to multiple primary dendrites that occupied a volume of roughly globular shape; they were bushy and spread out close to the soma. Their appearance was delicate in comparison to the dendrites of large MCs. There was a large variation in the density of neurites within the dendritic volume; while some cells ramified intensely, others were very sparse.

The small MC subclass stood out by making local, axonal synapses. The axon collaterals of small MCs extended in GL and PL and did not overlap with the glomerulus innervated by the dendrites. The local interactions were examined qualitatively by tracing along all their collaterals and inspecting synaptic partners for one small MCs of GL6, GL7 and GL9, respectively. One of these cells stood out by a high frequency of vesicles along the entire axon. Distinct in- and output synapses wer, however, restricted to the bulbar varicosities. The three MCs made axonal connections with a multitude of different INs (13, 26 and 47 connected neurites, respectively). Output synapses were more frequent than input synapses. Most connected partners were short neurite fragments that remained unidentified, many of which showed morphological features of the distal dendrites of DLINs and some PLINs (see below). Among the identifiable partners were INs from all layers of OB. These INs could innervate both the same and different glomeruli than the respective MC. Input was given to the dendrites of GLIN2s which also received input from dendritic synapses of the same small MCs. One MC gave input to the dendrites of a GLIN4. Several varicose PLIN3s and multiple DLIN2s of different subclasses received input or made reciprocal synapses with small MCs. Notably, none of the small MC axons received sensory input.

RCs

First described in the OB of goldfish by Kosaka & Hama in 1979, RCs have been reported in a multitude of other teleosts including zebrafish(Kosaka and Hama, 1979; Kosaka and Hama, 1980; Alonso et al., 1987; Arevalo et al., 1991; Fuller and Byrd, 2005). RCs were characterized by the namegiving ruff, a structure of extensive protrusions at the axon initial portion with reciprocal synapses to local INs. They were described to make altogether few synapses with other neurons, surround MC dendrites in a glial fashion and are expected not to receive OSN input(Kosaka and Hama, 1979; Kosaka, 1980; Kosaka and Hama, 1981).

RCs (11 FR, 8 CR, view) were generally well represented in the image volume. One cell, however, was cut right at the base of its axon and its ruff was not preserved in the imaged volume. It was identified as an RC by the morphology of its dendrite. All other fully reconstructed RCs were either completely represented in the stack or only minimally cut.

The somata of RCs were globular and of medium to large size (ø: 12 - 16 μm, $\bar{x} \pm \sigma$: 14.1 ± 1.4 μm). They were distributed in the GL and superficial PL. The surface was mostly smooth and bore only a few short, thin, sometimes sheath-like excrescences that were neither pre- nor postsynaptic. Cell bodies were usually positioned deep with respect to the glomerulus they innervated (Fig. 2B7). RC somata could be identified with high reliability from the outline and appearance of the perikaryon. Most characteristic was the ER, which formed concentric circles following the circular outline of the soma and created intricate, maze-like convolutions within the perikaryon. The RC somata were densely populated with cellular organelles. Each soma contained multiple Golgi apparatus, which were typically embedded between the interconnected chambers of the ER network. Endosomes and lysosomes were found, and the cytosol was filled with vesicles of various sizes and shapes. A myriad of mitochondria varying in size and shape was found in the soma. They were typically smaller in diameter than MC mitochondria. The light staining of the matrix contrasting the christae and a globular shape were characteristic for RC mitochondria.

Primary cilia were identified in all RCs with preserved soma. Most RCs (12/19) gave rise to two primary cilia. These emerged from the soma in immediate proximity to each other (Fig. 2B6) and were often parallel to each other. In one cell even a third cilium as observed. On average, RC cilia were the longest of all OB neuron classes (1.8 – 6.5 μm, $\bar{x} \pm \sigma$: 3.9 ± 1.3 μm). The cilia often bent towards the soma and took a course parallel to the cell body.

The axon generally arose at the pole facing the deep layers and opposing the base of the primary dendrite(s). It was usually small in diameter (ø: 0.5 – 1.25 μm, $\bar{x} \pm \sigma$: 0.9 ± 0.2 μm) and in contrast to MC axons the base was filled with ER and vesicles of all sizes. All axons were initially directed towards the deeper layers but then bent laterally and traversed the deep GL or superficial PL sometimes for more than 100 μm length. Three axons made short site branches locally, in which both inbound and outbound synapses could be detected. None of the axons were myelinated. In contrast to many other teleost species, in which the initial portion (IP) of the axon preceding the ruff may vary a lot in length(Kosaka and Hama, 1980; Alonso et al., 1987; Fuller and Byrd, 2005), the ruff was found to emerge within the first 5 μm of the axon, consistent with observations by Fuller & Byrd(Fuller and Byrd, 2005). All ruffs were positioned in the extra-glomerular neuropil of deep GL or superficial PL. One superficial RC had a very long dendritic stalk that projected towards the superficial neuropil of its glomerulus and only branched there.

The ruff had a cylindrical or conical outline and extended over a length of 25 to 35 μm on the axon (Fig. 2B). It consisted of multiple branchlike protrusions radiating all around the axon and typically ending in amorphous, varicose enlargements. With increasing distance to the soma, the ruff protrusions became shorter and sparsen out. The protrusions diverged frequently and intermingled strongly, thus densely filling the volume delineated by the ruff. With their variance in diameter and their bumpy surface these branches were reminiscent of MC dendrites. However, they did not possess any pointy or frizzy excrescences but showed rather rounded indentations and bulges. The cytosol of the ruff was filled with vacuoles, vesicles of various sizes, mitochondria, and ER christae with voluminous lumen. The central axonal strand could be made out because it accumulated large vesicles and was permeated by microtubules (Fig. 2B). Small and round synaptic vesicles filled the ruff protrusions and due to the small size of synaptic vesicle clouds, individual presynaptic sites were often hard to determine.

The dendritic tree of RCs was extensive and spanned a large volume. Due to the heavy branching combined with the delicacy of their branches, the outline of RC dendrites was reminiscent of a feather duster or besom. Primary dendrites emerged at the superficial pole of the RC soma. Most cells gave rise to a single large primary dendrite (ø: 1.0 – 4.5 µm, $\bar{x} \pm \sigma$: 2.3 ± 1.0 µm) that branched immediately at its base into several sub-branches. These main dendrites branched mainly asymmetrically and could thus be clearly distinguished throughout their entire length. They were of moderate diameter, <1 µm, and tapered off with distance to soma. They were characterized by frequent large vesicular membrane structures, ER, mitochondria and clearly visible microtubules. This was not seen in the finer branches and in the small twigs, which covered the shafts in large numbers. The finer branches were very delicate and had diameters of typically 100-200 nm. They meandered tortuously and often gave rise to small subbranches, but even more frequently to short twigs. At irregular intervals dendrites dilated into round bulbar enlargements, giving rise to a beaded appearance of the dendrite. These enlargements were densely packed with characteristically lightly stained mitochondria. Almost no cytosol surrounded these mitochondria. Usually, multiple sub-branches or twigs emerged from such a bulbar enlargement. The RC distal branches and twigs were characterized by a very clear cytosol; rarely small strands of ER and no vesicles were observed. In contrast to reports by Kosaka & Hama(Kosaka, 1980 #3018) no evidence of presynaptic structure on the RC dendrite was found.
The dendritic twigs were the most distinguishing feature of the RC dendrite, on average 54 ± 3% (n=8 FR) of a RCs skeletal length was contributed by twigs. They were thin and short, yet tortuous and typically ended in a rounded tip. In a subset of RCs, the shafts of bigger dendrites were completely covered by twigs resulting in a very uneven appearance while other RCs gave rise to twigs predominantly in their distal dendrite. The twigs could be postsynaptic to IN or sensory axons, but often, they wrapped around other neurites and squeezed into synapses between two other neurons. Occasionally, the twigs were observed to wrap intensely around each other, which gave rise to the peculiar structures.

Even in the most superficially positioned RCs the ruff extended into a volume of GL or PL neuropil, which was devoid of sensory axon terminals. The RC dendrite, in contrast, was fully immersed into a single glomerulus or glomerular compartment with only minor tip branches occasionally extending beyond. Early EM studies in goldfish suggested that RCs did not receive sensory input(Kosaka and Hama, 1982). Here, however, numerous synapses from sensory axons onto the RCs were identified. These targeted predominantly the thin, terminal dendritic branches. Sensory synapses to RCs were, however, far less prominent than the sensory inputs to MCs in both frequency and size.
An intriguing aspect of the RC-glomerular interaction was their interplay with MCs: The delicate RC dendrites grew alongside and heavily entwined with MC dendrites, wrapping their thin side branches around the larger MC neurites (Fig. 3). Reconstructing short dendritic branches Kosaka & Hama described the RC dendrite to coil around the MC dendrite. Looking at whole cell reconstructions of MCs and RCs provided insight into the frequency and intensity of this interaction: The RCs of glomerulus GL5 and GL7 contacted all MCs of their glomerulus or glomerular subcompartment. The intensity of the tight interaction varied substantially. In some cases, the RC followed the MC dendrite over several micron length tightly wrapping around MC dendrite. Sometimes the contact was limited

to a given varicosity of the MC dendrite. In a given MC-RC pair these contacts were usually observed across several distinct dendrite sections. By trend these anatomical alignments were more frequent between small MCs and RCs and more sporadic with large and intermediate MCs (Fig. 3B).
Despite this ample contact, synapses from the MC to the RC were sparse and often small. Instead, RC dendrites were often found in close proximity to MC synapses onto interneurons (Fig. 3A). RCs and MCs also received shared sensory and IN input (Fig. 3B).
RCs received substantial input from diverse INs of all layers on their dendritic tree. On the ruff RCs made many, often reciprocal, synapses with INs.

GLIN1

Morphology

The GLIN1s represented the smallest neurons observed and comprised 70 cells that can be viewed here. Their somata were found throughout the GL, though they appeared more frequent in the deeper GL and superficial PL. The majority of GLIN1s somata were positioned on or were part of a small cell cluster adjacent to a RC soma (55/69, ambiguous for one GLIN1 with CD3). Fewer somata lay freely in the neuropil near the ruff (14/69). GLIN1 somata were small (ø: 7 - 9 µm, $\bar{x} \pm \sigma$: 7.6 ± 0.5 µm) and ovoid or irregular in shape, rarely globular. Somata that were part of a cluster were usually flattened at one or more sides. The nucleus filled the perikaryon almost completely and often exhibited profound invaginations of the cytoplasm. Cytosol collected at the base of the primary dendrite(s) where cellular organelles like endosomes, lysosomes, and various large vesicles, the Golgi apparatus and few long strands of ER, as well as some small to medium-sized mitochondria could be found. The mitochondria resembled those observed in RCs: They were compact, roughly globular, or ovoid without branching and displayed a very light staining of the matrix. Mitochondria were occasionally also found in other parts of the cell body, squeezed into the small cytoplasm space between nucleus and plasma membrane. The surface of the soma was smooth and could bear small twigs and sheath-like excrescences. These excrescences squeezed in between the dense neuropil surrounding its soma and the ruff of an associated RC or wrapped around neurites and often received input from multiple neurites. No outbound synapses from the soma were observed. However, occasional inputs from neurites reminiscent of distal neurites of deep layer INs were found. Short primary cilia were found in 43 of 70 GLIN1s (0.2 – 4 µm, $\bar{x} \pm \sigma$: 1.4 ± 0.7 µm), eight of these gave also rise to a second nearby cilium. In eight cells without observable cilium, centrioles were found in the perikaryon.

Axons could be identified in most GLIN1s (68/70). In one cell the soma was cut at the image boundaries and in another no axon could be located. However, the latter cell displayed the characteristic intrusion of the ruff neuropil and the entanglement of the ruff and could therefore be identified as GLIN1. The axon usually originated from the soma as a very thin branch (ø: 100 – 500 nm, 63/68) or arose at any position along the dendrite (5/68). In nine GLIN1s the soma was positioned close to the stack boundaries and the axon was cut after few microns. In all other GLIN1s the axon was well-, in twenty-five GLIN1s fully preserved in the dataset. Axons made frequent collaterals which projected locally through the GL and in some cases, reached into superficial PL. The axonal projections appeared to be biased towards the volume occupied by a single glomerulus but could traverse up to three glomeruli ($\bar{x} \pm \sigma$: 1.1 ± 0.7 glomeruli). These glomeruli often but not always included the glomeruli that were innervated by the RCs to which a given GLIN1 was associated. By trend GLIN1s belonging to the same cluster and thereby associated RC occupied a shared volume in their axonal spread.
At irregular intervals GLIN1 axons displayed enlargements of non-uniform appearance. Most

neurons showed amorphous voluminous enlargements of variable length along the collaterals whereas in a subset rather flat and sheath-like enlargements dominated. The latter wrapped around other neurites, while the voluminous enlargements often exhibit grooves, in which neurites of other cells are imbedded. The varicosities typically contained one or more mitochondria and small, round synaptic vesicles were dispersed through the cytosol. The axonal enlargements were typically sites of synaptic interaction with other superficial and deep INs. These synapses were often reciprocal, but unidirectional pre- or post-synapses were also present. The voluminous enlargements were usually studded with one or several very thin (100-200 nm) extremities, sometimes long spiky protrusions, which protruded into the surrounding neuropil. Occasional synapses onto these twigs were observed.

The dendrites of the vast majority of GLIN1s spread directly adjacent to their soma. Only in three cells the dendrite emerged as a single stalk, which ramified only at some distance from the soma. The dendritic trees assumed a roughly ellipsoid or conoid shape with a cross section on the order of 15 – 40 µm (longer axis). The extent to which the dendritic branches filled this volume varied substantially. For the more prominent, denser, bushy versions, either a single dendrite that ramified immediately arose from the soma, or up to four dendrites emerged in close proximity of each other. The dendrites were already thin at their base (ø: 300 - 500 nm) and tapered off further to an average diameter between 100 and 200 nm. Similar to RC dendrites, short, meandrous twigs emerged from dendritic branches and sub-branches. Another similarity to RC dendrites exhibited by a subset of GLIN1s (20/70) were the bulbous enlargements containing mitochondria. However, these varicosities were far less frequent than in RCs. The dendritic cytosol contained occasional strands of ER, but appeared to be devoid of any vesicles, suggesting that the dendrite of these cells is purely postsynaptic.

Interactions With Other Neurons

Most characteristic for this IN class was their interaction with RC. GLIN1s were located adjacent to the ruff neuropil and immersed their dendrite fully into the ruff (Fig. 4C). The appearance of this interaction was reminiscent of creeper plant climbing a tree by attaching with their tendrils: The dendritic branches of GLIN1s were closely attached to the RC, entwining the ruff branches and protrusions by clasping the RC neurites with the small twigs. Despite this vast amount of contact sites, very few distinct synapses from the ruff onto the GLIN1 dendrite were identified. Often, the small twigs of GLIN1s protruded into the immediate vicinity of a synapse between a RC and another neurite. This made it difficult to distinguish whether RC vesicles next to the membrane contacted by GLIN1s formed an independent synapse or whether they belonged to the vesicle pool of the adjacent synapse with the other neurite. Yet, there were unambiguous instances of RC synapses onto GLIN1s, with usually more than three synapses per connected pair. These synapses were small with very few vesicles (Fig. 4C). Moreover, the interplay between RCs and GLIN1s extended to interactions with further neurons: GLIN1s frequently received inputs to their dendrites from neurites that were either reciprocally connected to the RC or made unidirectional in- or output synapses with it. Most of these neurites belonged to DLINs, e.g., inputs from DLIN1s and DLIN2s could be confirmed.

Dendro-dendritic connections between MCs and GLIN1s were improbable given the GLIN1 dendrites position outside of the glomerulus. To investigate potential axo-dendritic connectivity between GLIN1s and PNs, a pairwise screening for connections between GLIN1s and PNs of the glomerulus innervated by the GLIN1 axon was conducted. No synapses were found between GLIN1s and MCs (629 pairs). This screening revealed synapses from the GLIN1 axon onto the proximal RC dendrite in six of 121 inspected pairs. In five of these pairs the GLIN1 received input from the same RC to its dendrite.

The axons of a subset of GLIN1s (28/68) formed small bundles of two to five axons of *other GLIN1s or GLIN4s.* These bundles could involve INs innervating the same RC as well as INs interacting with different RCs and they could perpetuate for only few microns to several tens of microns of axon length. The engagement into such a bundle could occur at any distance from the soma. Oftentimes axonal varicosities of one axon wrapped around shafts or varicosities of other axons of the bundle. At some of these contact sites reciprocal or unidirectional synapses were identified including both connections between only GLIN1 axons as well as between GLIN1 and GLIN4 axons. Furthermore, reciprocal connections to the same passing neurites were found from axons in the same bundle. Some GLIN1s provided sparse input to GLIN3s. GLIN1s were also found to synapse onto PLIN3s and PLIN4s as well as with various DLINs. These connections could be reciprocal. PLIN1s, an IN class that also stands out by its intense interaction with RCs, were found to provide input to GLIN1 dendrites through multiple synapses.

GLIN2

Morphology

The 30 GLIN2s comprised the largest of the INs of GL (view). The somata of GLIN2s were loosely distributed in the neuropil of the GL, and could be found at the bulbar surface, and in the very superficial PL. Occasionally, they clustered with one to three somata of various cell types (11/30). GLIN2 somata were globular to ovoid and medium-sized (ø: 8 - 11 µm, $\bar{x} \pm \sigma$: 9.8 ± 0.7 µm). The nuclei often had an undulated outline with mild indentation and were surrounded by a moderate, sometimes small amount of cytosol. Altogether, the perikaryon had a rather light and clear appearance. Some mitochondria were found distributed around the soma with a larger frequency at the root of primary dendrites. The mitochondria were remarkably small in diameter and overall size and stood out by a very darkly stained intermembrane space from which the cristae barely stood out. Few, rather short strands of ER cisternae surround the nucleus and relatively few, large vesicles and structures of the endo-membrane system were observed. The cells contained up to three Golgi apparatus, which could be very compact, but on other occasions stretched over a considerable length into the primary dendrites. The surface of the cell bodies was smooth but usually bore a few thin, filopodial excrescences. These were occasionally targeted by synapses. Direct input to the soma was also commonly observed, but no outbound synapses could be found. A primary cilium emerged at the base of a larger dendrite (0.5 – 6.0 µm, $\bar{x} \pm \sigma$: 3.4 ± 1.5 µm). In three cells this cilium projected and merged back into the soma.

GLIN2 axons emerged from the soma (14/25) or dendrite (11/26) and were identified as the only neurite with outbound synapses. These synapses were confined to bulbar or voluminous enlargements which appeared at variable distance from the axon origin. Among the five GLIN2s without identified axons, two cell bodies resided close to the stack boundary, such that the characteristic bulbar enlargements carrying outbound synapses could not be unambiguously identified as axon. In the other three cells, no axon was identified despite targeted search. A possible explanation is that their neurites often became extremely thin, which complicated segmentation and reconstruction. In GLIN2s with dendritic axon origin, the axon could emerge from a proximal dendrite, as well as from a dendrite of higher branch order. GLIN2s located in the deeper GL usually projected to the deep layers, while the axons of superficial neurons were directed predominantly laterally in GL. Axon collaterals were identified in nine GLIN2s. No axon was represented fully in the

stack. However, the axons of three GLIN2s which were better preserved traversed large volumes of the GL and PL but remained unmyelinated.
The bulbar, voluminous axonal varicosities varied considerably in frequency and size among neurons and had smooth appearance without excrescences. In between varicosities the axon was often constricted to a diameter below 200 nm giving rise to the appearance of a beaded chain (Fig. 5B). Larger varicosities could contain a single compact mitochondrion and make more than one outbound synapse. Even in enlargements with only one presynaptic site, vesicle clouds were large. Some varicosities had postsynaptic sites which could be part of a reciprocal synapse. Synapses onto the thin axonal shafts were not observed. Synaptic vesicles were exclusively found in axonal varicosities. There appeared to be a bimodal distribution of vesicle size: larger round vesicles with visible light lumen and small, dot-like vesicle. Synapses with only one kind of vesicle but also with both types of vesicles in the vesicle cloud of a given synapse were observed.

Up to four dendritic branches could emerge from the soma. Most GLIN2s had one or two large dendrites with a diameter of 0.8 – 3.5 µm ($\bar{x} \pm \sigma$: 1.5 ± 0.7 µm) and additional smaller dendrites, sometimes as fine as 150 to 500 nm in diameter ($\bar{x} \pm \sigma$: 380 ± 110 nm) that resembled subbranches of the main branches. In three GLIN2s of overall smaller size, large dendrites were missing, instead four smaller dendrites emerged. The larger primary dendrites gave rise to many subbranches. The path length from the soma at which the larger primary dendrite started branching varied significantly. While the primary dendrites in some cells branched immediately at the soma, others extended a dendrite stalk up to ~90 µm from the soma before ramifying. In most cells this stalk had a length between 15 and 30 µm.
GLIN2s were present in all glomeruli and even in this non-exhaustive reconstruction larger glomeruli were found to encompass multiple GLIN2s. The dendritic trees were entirely confined to the GL and the bulk of the dendrites was restricted to a single glomerulus or glomerular subcompartment with only a few distal branches protruding into neighboring extraglomerular neuropil. In 24 GLIN2s a minor, but unneglectable dendritic branch targeted regions beyond the glomerulus innervated by the main portion of the dendrite. These dendrites could originate as one of the thinner branches from the soma. When these side branches did not project outside the imaged volume, they extended into neuropil not occupied by any glomeruli and made minor ramifications.

The overall shape of the dendritic trees varied considerably between cells (Fig. 5A). The final outline of the dendrite seemed to be predominantly dictated by the shape of the primary glomerulus they innervate and the position of the soma. The more superficial representatives were often positioned laterally to their main glomerulus, sometimes at considerable distance. The dendrites of these cells adopted a fan- or tassel-like outline, sometimes more reminiscent of a flattened besom. Further inside the OB, GLIN2s were positioned at the deep side of their primary glomerulus. Their dendrites grew into a circular fashion, resulting in a donut-like shape. Despite the frequent and wide branching, the dendrite filled the volume it occupied only to a moderate extent. The branching pattern was asymmetric: primary dendrites were clearly discernible over a considerable length and tapered off slowly, while the emerging branches were as thin as the small dendrites arising from the soma. The dendrites were round in cross section and smooth in surface. Both larger and thin dendrites made regularly occurring dilations giving rise to a sparsely beaded appearance. These varicosities could contain slim and elongated mitochondria with a characteristically darkly stained matrix from which the densely packed cisternae barely stood out hat. Oftentimes these mitochondria obtained a twisted and curled shape. Within the GL neuropil GLIN2 dendrites stood out by their light appearance. Besides occasional mitochondria only darkly stained ER was found in long strands throughout the entire length of the dendrites, thick and thin, with the filiform processes at the appendages (see below) as the only exception. Occasionally larger vesicular structures with clear lumen could accumulate in more distal branches. But no synaptic vesicles were observed anywhere in the dendrite, suggesting it was a purely postsynaptic structure.

The dendrites were round in cross section. Their surface was mainly smooth but on the proximal part the dendrites were sparsely studded with protrusions which resembled small stumps. The thin dendrites were covered with hand-like appendages, an identifying characteristic for this neuron class. On the larger dendrites, they became increasingly more frequent as the distance from the soma increased. In the thin side branches, these could emerge directly from the dendrite. On the thicker dendrites, the filiform appendages were similar to sessile spines directly attached to the shaft, but they mostly occurred at the end of a thin filopodial spine neck. The length of these varied considerably, from a few hundred nanometers to several microns. The neck could be extremely thin and was many times too thin to be segmented. Such thin protrusions were easy to miss and more likely to be lost during neuron reconstruction, since these appendages were major input sites, this may affect connectivity measure.

The long necks terminated in a palmate, voluminous enlargement that carried multiple thin filiform extensions. The arrangement was reminiscent of long-fingered hands clutching around the presynaptic partner(s): The filiform processes enclosed around one or more bypassing neurites, which synapsed onto the voluminous enlargements. Not all "fingers" clasped around other neurites, some merely stretched out into the surrounding neuropil. They were not observed to be targeted by synapses. Synapses were directed towards the bulky "palm", which always contained darkly stained ER and, when larger, also one or more small darkly stained mitochondria. MCs presented a major input to these appendages, however some inputs from reconstructed INs were also observed see below.

Interactions With Other Neurons

The connectivity between GLIN2s and PNs of the OB was examined for three GLIN2s innervating GL7 and two GLIN2s projecting to glomerulus GL6. In GL7 GLIN2s projected only to the smaller subcompartment and received input from all eight MCs innervating this subcompartment. For two of these INs the number of contacts was remarkably high, and a given MC-GLIN2 pair could be connected by more than ten synapses. These synapses were distributed widely across the dendrites of the connected pair. They could originate from the thick primary dendrite of MCs, but more often originated from their distal varicosities, and usually targeted the filiform appendages of the GLIN2 and to a lesser extent the enlargements of the main GLIN2 dendrite. Input to the third neuron was less frequent but still exceeded four synapses seven of eight pairs. The GL6 did not show compartmentalization and both GLIN2s received input from a large fraction of MCs (9 of 12 (76%), 6 of 12 (50%), respectively). While one of these GLIN2s was connected to MCs by multiple synapses per pair, the second GLIN2 was only connected through one or two synapses per pair. Altogether, these observations indicate that MCs contribute strong input to GLIN2s.

The dendrites of GLIN2s and RCs overlapped heavily within the glomerular neuropil. RC dendrites seemed to follow the projection of the GLIN2 dendrites and approach particularly in the vicinity of input synapse from MCs, but showed no tight entanglement as observed between RC and MC. No synapses between GLIN2s and RC dendrites were observed. A separate pairwise overlay of 19 GLIN2s with 17 FR and CR RCs as well as additional 17 RCs, which were not reconstructed and predominantly represented as a ruff protruding into the imaged volume, yielded nine contact sites between a ruff and GLIN2 dendrites and four contacts between a GLIN2 axon and a ruff, but no contacts between a GLIN2 axon and RC dendrites. At two of the contact sites a synapse from an RC onto two different GLIN2 dendrites was identified.

During neuron reconstructions various interactions between GLIN2s and local INs were observed. Regular inputs originated from GLIN3s and PLIN2s innervating the same glomerulus and targeted predominantly varicosities on the dendritic shaft or sessile appendages. Multiple GLIN2s received inputs from PLIN1s that made synapses onto either the filiform appendages or the dendritic shaft. Several DLIN1_ls were found to provide input to GLIN3.
One conspicuous type of input originated from an unknown source: Thin neurites entering the imaged volume mostly in PL or GL formed amorphous, varicose enlargements at irregular intervals that were filled with small granular synaptic vesicles. In contacts with GLIN2s, large, usually sessile GLIN2 appendages tightly enveloped the neurite varicosity. The filiform extensions not only enclosed the varicosity but often protruded into the bulk of the varicosity. Multiple synaptic active zoned could be identified. The emerging structure could be outstandingly large and could involve multiple GLIN2s as well as multiple presynaptic neurites (Fig. 5B3). Regularly, these inputs could be found on the secondary GLIN2 side branch that innervated extra-glomerular neuropil. Some neurites were also found to (additionally) synapse onto the GLIN2 soma or sessile, filiform appendages on the proximal dendrite. These contacts were usually less conspicuous.
None of the presynaptic neurites could be traced back to a soma. While these neurites might presumably belong to DLINs with cell bodies outside the imaged volume, they differed distinctly from the distal dendrites of identified DLINs. The varicosities were mostly larger than the usual distal DLIN varicosity and less frequent and no sheath-like enlargements were observed. Branching was common but the branching angle was often wider than typical for DLIN distal dendrites. Further, reciprocal connections to MC which are abundant in DLIN dendrites were not observed. Aside from the contacts with the GLIN2s these neurites made synapses with other local IN dendrites and somata, namely PLIN3s. The vertebrate OB is known to receive abundant top-down projections(Burton, 2017 #2751), it may be possible that these neurites represented centrifugal axons.

None of the GLIN2 axons were completely represented within the imaged volume. A random sampling showed these axonal varicosities were predominantly presynaptic but also revealed the presence of inbound synapses, some of which were part of reciprocal synapses. In this context, two synapses from a GLIN2 onto a large MC dendrite were observed. The GLIN2s gave input to a variety of descending neurites of DLINs of different morphological classes predominantly targeting the descending main dendrite. As such, several DLIN1s with long-necked spines were found to receive input from GLIN2s. Furthermore, outputs onto the proximal dendrite or somata of GLIN3s innervating different glomeruli were found. Other postsynaptic partners included axons of unknown origin some similar to GLIN2 axons themselves; others with larger, less bulbar rather amorphous varicosities at lower frequency. Some of these synapses were reciprocal.

GLIN3

Morphology

Most of the 18 GLIN3s were well preserved in the imaged volume (view). The overall size of GLIN3s varied substantially (Fig. 6A). The GLIN3 cell bodies were located in the deep GL and superficial PL. All somata were part of small cell clusters with heterogeneous composition of cell types, which could involve other INs, glia or more rarely a MC. The surface of the somata was flattened where it touched other cell bodies, and was otherwise bumpy and uneven, but not ragged. It was populated to varying degrees with filopodial dendritic twigs and a few protrusions that resembled branched or long-necked spines but were commonly not recipients of synaptic input. The somata were of small to medium size (ø: 8 - 10 µm, $\bar{x} \pm \sigma$: 8.8 ± 0.6 µm), roughly ovoid and contained little or moderate amounts of cytosol. The nucleus filled the soma almost completely and followed the outline of the soma in shape. Except for the base of the biggest primary dendrite, only few short strands of ER and very few slender mitochondria were visible. A single primary cilium of medium length arose from each soma (1.5 – 4.5 µm, $\bar{x} \pm \sigma$: 2.4 ± 0.7 µm). No vesicles and no outbound synapses from the soma were observed. At the base of the biggest dendrites mitochondria in larger numbers and ER accumulated and the single Golgi apparatus usually stretched into the dendrite for a couple of microns.

Despite the absence of an axon different parts of the GLIN3 dendrites exhibited differential in- and output properties. The dendritic branches could be classified into two different morphological groups: a larger, superficially projecting type and a delicate type. The latter bore many branched or long-necked spine-like appendages and ramified predominantly laterally in the deep GL and superficial PL. The thin dendrites could arise from the soma or the proximal parts of the larger dendrites. Their diameter was usually smaller than 500 nm. Number and length of these dendrites varied a lot between cells: Some cells bore multiple shorter branches with a length up to 20 µm. Others had only one or two thin dendrites, which extended for far more than 50 µm and made few side branches. Few of these longer dendrites were projected towards the superficial GL, where they ramified and showed comparable morphological appearance as the distal portions of the larger dendrites (see below). The thin dendrites were covered with a multitude of protrusions and small side-branches bearing spine-like appendages of diverse shapes. This gave rise to the hairy appearance of these neurons (Fig. 6A). Some protrusions had a rather typical morphology of thin spine-like appendages with a small head, which was often flattened in one axis, and a usually rather long neck (>1.5 µm). Few branched and short-necked as well as filiform spine-like appendages were observed. Filopodial branches without spine head were also very frequent. A random number of spine heads, filopodial twigs and side branches was inspected in numerous GLIN3 and the longer thin dendrite along the entire length for two neurons. All spine heads received an inbound synapse. Input to the filopodial protrusions was less frequent and occurred at all positions along the protrusion. A considerable number of synapses onto the main dendritic shaft were observed, but no presynaptic structures.

The soma gave rise to one, sometimes two main dendrites, which at their base had a diameter between 0.5 and 1.5 µm ($\bar{x} \pm \sigma$: 1.1 ± 0.4 µm). The outline of the cross section was irregular: It was flattened in one axis, thus, rather oblong, and varied frequently in the ratio of the two principal axes. Short, darkly stained strands of ER stretched throughout the dendrite. Microtubuli were clearly discernible. The proximal portions of the dendrite were covered with spin-like, filopodial protrusions and thin branches like the proximal dendrites. In larger GLIN3s these dendrites were projected towards the superficial parts of the GL and could obtain considerable length (> 100 µm). However, shorter stalks were also common. The dendritic stalk made a few, very irregularly shaped dilations that contained lots of ER and sometimes few, small mitochondria immersed in large cytosolic volumes. No synapses were observed at these locations (Fig. 6B). At their distal end the larger dendrites ramified increasingly and formed a besom-like, cone-shaped, or bushy tree. In most cells the frequency of spine-like appendages and filopodia on the thicker dendrite reduced gradually towards the distal parts. In smaller representatives the dendritic stalk was much shorter (<10 – ~20 µm) and the transition between the appearances of the proximal to the peripheral dendrite occurred in relatively close proximity to the soma. The dendritic ramifications resulted in a more globular, bushy outline of the cell.

In the peripheral dendritic tree, spine-like appendages were absent. Instead, this part of the dendrite bore enlargements and protrusions of a different appearance: irregularly shaped, sometimes voluminous enlargements, which showed indentations and concavities formed by contacting other neurites, emerged at high frequency all over the distal dendrite (Fig. 2B2). They could sit on branch points, as dilation of a dendritic shaft, like sessile spines on a branch, or similar to mushroom spines at the end of a neck-like branch. Most notably, the vast majority of these structures contained synaptic vesicles and were not only sites of inbound, but also outbound synapses (Fig. 2B2), which could be unidirectional or reciprocal. As common for sites of synaptic output, these enlargements often contained the small, slender mitochondria characteristic for GLIN3s surrounded by ample cytoplasm.
In many neurons the change from proximal/deep and distal/superficial dendritic morphology could be pinpointed to a particular bifurcation, arising at one of the enlargements of the dendritic stalk described above (Fig. 2B). Other GLIN3s did not show such a clear-cut transition, here the distal dendrite appeared more widely spread, the enlargements were less dense and more often sat at the end of a neck. Nevertheless, a qualitative examination suggests presynapses were restricted to the distal dendritic enlargements in these GLIN3s as well.

The thin proximal dendrites resided outside the volume in which sensory axons terminated. The emergence of the morphological phenotype of the distal dendrites coincided with the entrance of the dendrite into a glomerulus. The majority of GLIN3s innervated a single glomerulus or glomerular subcompartment. Secondary branches that traversed into the glomerular neuropil were observed in nine GLIN3s. In four of these these dendrites terminated and ramified in the same glomerulus as the main dendrite. Three GLIN3s innervated a second glomerulus and connected to PNs in there. In another two GLIN3s the second dendritic branch was projected towards a volume remote from the glomerulus innervated by the primary dendrite, but outside the imaged volume.

Interactions With Other Neurons

Synapse detection in GLIN3s was complicated by the vesicles loosely scattered in the cytosol of the enlargements of distal dendrites. Furthermore, output synapses could be rather small, comprising only a few vesicles associated with the membrane. Nevertheless, distinct synapses were detectable.

The interactions with the PNs for three GLIN3s in glomerulus GL6 and one GLIN3 in the glomeruli GL12, GL11 and GL7, respectively. Most GLIN3s broadly connected to MCs of the glomerulus or glomerular subcompartment they projected (ID414: 4/8 MCs (50 %), ID987: 7/12 MCs (58 %), ID1330: 3/12 MCs (25 %), ID1008: 6/7 MCs (86 %), ID1346: 5/12 MCs (42 %), ID2868: 4/12 MCs (33 %)). Usually each connected MC-IN pair made multiple contact sites, in which an enlargement of the GLIN3 dendrite was either wrapped around or immersed into the MC dendrite. This interaction occurred predominantly at the thicker MC dendrites and to a lesser extent at their distal varicosities. The number of contact sites between pairs varied from one to far more than ten contact sites and all inspected contact sites contained synapses. Most pairs were connected through more than three synapses. The majority of MC-GLIN3 pairs were reciprocally connected (23/29). In the 23 reciprocally connected pairs most synapses were reciprocal, however, additional unidirectional synapses, both in- and outbound, were observed in 19 pairs. In four pairs a MC provided input to a GLIN3, and the reverse connection was found in two pairs. The three GLIN3s innervating GL6 strongly overlapped: They formed localized dendrite bundles, made reciprocal connections with the

same MC in close proximity and received shared input with that MC from passing neurites. One of the GLIN3s projected a smaller dendrite to GL3, where it also connected reciprocally to a MC.
Among the 16 inspected RC-GLIN3 pairs three were found to be connected with a single GLIN3 giving input to the dendrites of all three RCs of GL7 through one or several synapses. During the reconstruction process an additional one more GLIN3-RC pair in which the GLIN3 was presynaptic was observed and another pair in which the GLIN3 received several inputs to the spine-like appendages of the proximal dendrite.

Connections detected during reconstruction and the inspection of the proximal dendrites provide some insight into the interaction of GLIN3s with the IN network of OB. This showed ample input originating from passing neurites onto somata, spine-like appendages and proximal dendrites of GLIN3s. In most cases the source of these inputs could not be determined. These neurites were usually very thin (ø: 100 - 200 nm). Their synapses were located on small enlargements, which assumed shapes varying from bulbous to flattened trapezoids, and bore indentations and bulges, where they contacted other neurites. The enlargements were often, but not always, packed with small dot-like vesicles. No overlap with the neurites providing input to the large GLIN2 appendages was noted.
Some of inputs to the proximal GLIN3 dendrites and spine-like appendages showed morphological features reminiscent of PLIN3s, others resembled the distal branches of DLINs. Indeed, connections with DLINs of all subclasses were found in which the GLIN3 received input and one connection was found in which a GLIN3 provided input to a DLIN1_l. Both input connections from reconstructed DLIN2 and PLIN3 were observed. GLIN3s were further found to receive input from GLIN1s, GLIN2s, PLIN2s and other GLIN3s. Eleven connections were found in which the GLIN3s gave input to dendrite shafts of the GLIN2s, and one outbound connection to a GLIN4.

GLIN4

Morphology

The axon-bearing GLIN4s were represented by 16 reconstructed neurons (view). Their somata were distributed from the very superficial GL to central PL. The somata of six GLIN4s were part of a small cell cluster, four of these were directly attached to an RC soma. GLIN4 somata were small (ø: 7 - 9 µm, $\bar{x} \pm \sigma$: 8.5 ± 0.6 µm) and their shape ranged from roughly globular to ovoid. All somata were enveloped by a glial sheath to vast parts if not entirely. They rarely bore any excrescences. The nuclei had a very irregular outline and were surrounded by only a thin layer of cytosol. A single Golgi apparatus was located at the base of the dendrite where cytosol was more abundant. Here, other organelles of the endo-membrane system could be found. Mitochondria could be large and branched and the lightly stained matrix dominated their appearance. The mitochondrial cristae did not arrange densely packed lamellar stacks but formed a loose, irregular netting with low to moderate density in the matrix. Whilst overall lumen size was variable, an abundance of large, vesicle-like cristae with clear intermembrane space was observed. Short neuronal cilia emerged from the base of the dendrite in 12/16 GLIN4 with a second cilium identifiable in four of these cells (length: 0.5 – 1.5 µm, $\bar{x} \pm \sigma$: 0.8 ± 0.3 µm). Another GLIN4 had formed two centrioles without discernible neuronal cilium.
Axons arose from the soma (12/15) or dendrite (3/15) with a diameter < 500nm. In one cell no axon could be identified. Based on congruence in other morphological features this cell was classified as GLIN4. Given the delicacy of GLIN4 axons it is possible the axon was detached during the sample preparation or image acquisition, or it may have been missed in the reconstruction process. GLIN4 axons resembled GLIN1 axons with which they could form small local bundles. They gave rise to local

collaterals and made amorphous, varicose enlargements which contained pre- but also postsynaptic sites, including reciprocal synapses. GLIN4 axons were well represented within the image volume, but only two were completely contained. The axonal projections of the more superficial GLIN4 remained in GL, whereas the axons of deeper GLIN4s ones spread towards or through PL. Most GLIN4s projected some portion of their axon through the neuropil volume of one or two glomeruli ($\bar{x} \pm \sigma$: 1.5 ± 0.5 glomeruli). However, five GLIN4 axons followed a trajectory that remained in the neuropil inbetween glomeruli.

The soma gave rise to a thin, single dendrite (ø: 0.3 - 1 µm), which continued as a stalk for few to several tenth of microns before ramifying into a small, often sparse dendritic bush that span a volume of ca. 15-50 µm in diameter. Except for two GLIN4s innervating two glomeruli in approximately equal portions, GLIN4 dendrites were restricted to a single glomerulus or glomerular subcompartment. GLIN4s were identified in seven of the twelve glomeruli. The dendrite shared morphological features of the GLIN1 dendrite: The branches were delicate with a diameter usually smaller than 300 nm. The dendrites often widened at branch points where one or two small, globular mitochondria of similar appearance to the somatic mitochondria could be found. This gave rise to a beaded appearance of the dendrite in six GLIN4s. Apart from mitochondria darkly stained ER could be traced through the dendrites. No vesicles could be found. Tip branches constituted a large portion of the dendritic tree.

Interactions With Other Neurons

Ten GLIN4s were found to tightly wrap their dendrites around RC dendrites in a fashion similar to the interaction between MCs and RCs: The smaller GLIN4 dendrite was most closely attached to the bigger RC dendrite (and sometimes soma, Fig. 7C). The dendro-dendritic interaction spanned over varying portions of the IN dendrites but was altogether less pronounced than the interaction between RCs and GLIN1s. Since the dendrites of both GLIN4s and RCs are presumably purely post-synaptic structures, this suggests that the GLIN4s tracks the input to the RC. Indeed, several instances of synaptic inputs from a given neurite onto both RC and GLIN4 were observed (Fig. 7C).

The connectivity of PNs and GLIN4 was further investigated by pairwise inspection of GLIN4 and PNs of the glomeruli innervated by GLIN4 dendrites and axon. Of the 330 GLIN4-MC pairs four MC-to-GLIN4 and two GLIN4-to-MC pairs were found. These pairs were connected by one or two synapses. Among the 56 RC-GLIN4 pairs one pair reciprocally connected through reciprocal synapses between GLIN4 axon and ruff was identified. In addition, one synapse from the ruff of an RC onto a GLIN4 axon was identified during the process of neuron reconstruction.

Further, both reciprocal and inbound connections with GLIN1s and DLIN1s, GLIN4-to-PLIN4s and PLIN4-to-GLIN4 connections as well as input connections to GLIN4s from GLIN3s, DLIN2s and DLIN3s were detected.

PLIN1

Morphology

The PLIN class I (PLIN1, view) comprised a group of six small, bushy neurons that were almost entirely confined to the PL and generally well preserved in the stack. The small somata were of roughly globular shape and embedded in the PL neuropil or part of a deep layer clusters (ø: 7 - 9 µm, $\bar{x} \pm \sigma$: 7.8 ± 0.9 µm). The somata had a rough surface and bore membrane excrescences that did not resemble spines. The contour of the nucleus roughly followed the soma outline. Only a thin layer of

cytosol separated it from the plasma membrane. Cellular organelles resided in a cytoplasmic protuberance at a pole of the soma. In two cells this protuberance was formed as a bulge, which was connected to the soma via a short bridge and gave rise to the primary dendrite. A single, short primary cilium emerged from either soma or the cytosol bulge at the base of the dendrite (1.5 – 2.5 µm, $\bar{x} \pm \sigma$: 2.0 ± 0.3 µm)
One or two small to medium-sized dendrites arose from the somata (ø: 0.5 – 1.0 µm, $\bar{x} \pm \sigma$: 0.6 ± 0.2 µm). The dendrites were projected towards the deep GL or superficial PL, where they ramified into a bushy dendritic tree with a diameter on the order of 20 to 30 µm. In addition, the PLIN1s bore several thin, proximal branches of variable length, sometimes reminiscent of long-necked spines, which emanated from the soma or arose from the main dendrite. These proximal branches made small, infrequent, flattened rather than voluminous varicosities, which could occur at any position on the branch. In one neuron multiple thin branches emanated from the deep anterior pole of the soma, forming a sparse version of the distal, bushy ramification. This ramification, like the distal dendritic trees of PLIN1s was immersed into RC ruffs and where synaptic contacts with the RC were made. The distal dendrite exhibited varicosities at a much higher density. Like the varicosities in other parts of the dendrite, they were predominantly amorphous and flattened, but bulky voluminous enlargements, usually containing mitochondria, were also observed. The distal dendrite also made many thin, almost pointy tip branches. Some of these tip branches spread far outside the main globular dendritic ramification.

A random subset of proximal and distal branches was screened for synapses. No clear-cut pre- and post-synaptic compartmentalization in these neurons could be determined. Presynapses were small, with vesicle clouds comprising usually only a handful of vesicles. Pre- and postsynaptic structures were most frequent in the distal dendritic varicosities but were not limited to these structures and occurred on shafts as well. Notably, not all varicosities made outbound synapses.

Interactions With Other Neurons

The PLIN1s showed little interaction within glomeruli. Even the most superficially projecting neuron spread its ramification in a cleft between the GL9 and the GL7 glomerulus, which was not innervated by any MCs. Three neurons projected minor parts of the distal dendrites into a single glomerular volume. Two of these projected to GL5 and made connections with one or two out of four MCs, respectively. These connections were mediated by one or two reciprocal synapses with four additional synapses onto the MC in one pair. The other cell made no synaptic contacts to PNs.

PLIN1s stood out by a very tight association with RCs. Unlike GLIN1s, they could be associated with multiple RCs (1-5, $\bar{x} \pm \sigma$: 3.2 ± 1.3 connected RC/PLIN1). The PLIN1s contacted the RCs not only via the distal, bushy dendrite, but also any other of their side branches. The fraction of dendrite projected into a given ruff varied significantly: Most PLIN1s immersed their bushy distal dendrite into one or at most two RCs and projected only a few smaller branches into additional RC ruffs. In some of these smaller projections no synapse with the RC was observed. Connections via the bushy distal dendrites involved several reciprocal, but also additional unidirectional in- and outbound synapses with two to five synapses per pair. In addition to the 13 reciprocally connected RC-PLIN1 pairs, in four of the 19 pairs the RC gave input to the PLIN1 and in two pairs the PLIN1s provided input to the RC dendrite. The unidirectionally connected pairs encompassed one or two synapses.

Synapses from PLIN1s to other neurites within the volume of a ruff were observed. To explore the connection between PLIN1s to GLIN1s one PLIN1 was overlayed with all GLIN1s associated to the same RCs and screened for contacts. The PLIN1 gave input to five of ten reconstructed GLIN1s

innervating the same three RC ruffs. The number of contact sites varied between one and four synapses but was not necessarily restricted to a single ruff if both pairs innervated the same second ruff.
In addition, several PLIN1s were found to provide input to different GLIN2s, these connections consisted of a single synapse each.

PLIN2

Morphology

PLIN2s were represented by a small group of 11 axonless INs which were characterized by a thin, elongated main dendrite, which penetrated the glomerular area (view). The somata of these INs occurred as members of pairs or small heterogeneous clusters in the PL and had a small diameter (ø: 7 - 9 µm, $\bar{x} \pm \sigma$: 8.0 ± 0.6 µm). They were roughly globular with a rough surface which in a subset of PLIN2s bore very few amorphous membrane excrescences or spine-like appendages. Typically, somata showed several indentations made by larger neurites or neighboring somata. The perikaryon was only represented by a thin layer of cytosol separating the nucleus from the plasma membrane, and the indentations of the outer membrane were sustained in the nucleus outline. Very few long and thin mitochondria and basically no ER cisternae were found in the cytosol surrounding the nucleus. The bulk of cellular organelles collected in protuberances at the base of the primary dendrite or were fully outsourced to proximal parts of that dendrite. Each PLIN2 possessed a short primary cilium (1.3 – 2.0 µm, $\bar{x} \pm \sigma$: 1.7 ± 0.3 µm).

Most PLIN2s gave rise to a single main dendrite, only one cell gave rise to a second dendrite. The dendrite was medium-sized in diameter (ø: 0.75 – 1.25 µm, $\bar{x} \pm \sigma$: 0.92 ± 0.24 µm). Maintaining a constant circumference, it traversed the PL in a nearly straight path without branching, while its oblong cross-section slowly rotated. The main shaft was garnished with protrusions that occurred at quite regular intervals on the order of several microns. The length of this interval as well as the appearance of the protrusions varied between cells. The appearance of the protrusions varied substantially: In some cells the protrusions were short studs of variable shape some reminiscent of sessile spines with a saw-tooth-like appearance. In some PLIN2s protrusions showed a mixture of filopodial, long-necked and long branched spine-like appendages, whereas in other PLIN2s the appendages rather resembled flattened, round headed spines. Protrusions were randomly visited in a few cells. Some but not all received synaptic input which originated from varicosities on thin neurites that contained many small, dark-stained, dot-like, synaptic vesicles (Fig. 9B) and could not be traced back to any local cell type. In three PLIN2s fine dendrites of a few microns length branched from the soma or proximal dendrites.
When entering the glomerular volume, the dendrite branched and formed a besom-shaped tree with thin branches (ø: 200-300 nm). Dendrites remained a given glomerulus and often occupied only a subsection of the glomerulus or glomerular subcompartment. With two exceptions the dendrites were sparse and showed little branching. This distal dendrite did not bear spines but exhibited amorphous varicosities that contained synaptic vesicles. Flattened and bulky varicosities occurred in equal measure in each cell. They could be located along the branch, but more often occurred at tips and, thus, gave an appearance of spines with a very bulky head.
PLIN2s were morphologically close to the GLIN3s. They set themselves apart from this group by their lack of proximal dendrites, the regularity of their spine-like protrusions and the predominant sparseness of their distal dendrite.

Interactions With Other Neurons

Outputs of PLIN2s were only observed within the neuropil of the glomerulus they innervated. The connectivity of PLIN2s with PNs was investigated for four PLIN2s innervating the glomeruli GL5, GL6 and GL5. The PLIN2s of GL6, GL5 and GL8 connected broadly to the MCs (ID1104: 6/12 MCs (50 %), ID2479: 3/5 MCs (60 %), ID2980: 4/9 (44%)). The PLIN2 innervating GL9 made connections with only 15 % of the MCs in the glomerulus (2/13 MCs). While this cell had a rather sparse dendrite that only projected to a subvolume of one of the two subcompartments GL9 it seemed to be selective among MCs innervating this neuropil region.

Most PLIN2-MC pairs were reciprocally connected through reciprocal synapses (10/13 pairs). The number of synapses within a given pair could vary substantially. Three pairs were connected through a single reciprocal synapse. The more densely connected other seven pairs also made additional unidirectional synapses in both directions. The number of synapses per pair could be very high with up to 25 synaptic contacts per pair, respectively. Three MC-to-PLIN2 and two PLIN2-to-MC connections, connected through a single synapse each, were found. No connections with RCs were found.

The interaction with other INs of the OB was not explored in detail and during the reconstruction process only very few connections to reconstructed partners were observed. A synapse from a GLIN2 onto the proximal shaft of a PLIN2 was found and two pairs in which the PLIN2 provided input to the dendrite of a GLIN2. In addition, synapses from a PLIN2 onto a GLIN3, a PLIN3 and a PLIN4 were found.

PLIN3

The 19 axonless PLIN3s were characterized by usually a single, long dendrite that spread widely in a swirl-like fashion through GL (view). While sharing an overall coarse morphology, subtle differences in morphological details which were largely correlated across several features allowed to separate PLIN3s into two subclasses, ribbon-like (PLIN3_rl) and varicose PLIN3 (PLIN3_v).

Morphology

The PLIN3 somata were ovoid-shaped, of medium size (ø: 8 - 10 µm, $\bar{x} \pm \sigma$: 9.3 ± 0.7 µm) and could be found throughout PL. The ovoid nuclei were encompassed by little cytosol which contained few, very small mitochondria, or narrow mitochondrial side branches. Few, large vesicles, and short strands of ER were observed. The elongated ovoid shape of the somata were emphasized by a large funnel-shaped bulge which formed at the pole giving rise to the dendrite. Here, large amounts of cytosol and the elements of the endomembrane system accumulated, and a primary cilium of medium length emerged (2.0 – 3.5 µm, $\bar{x} \pm \sigma$: 2.4 ± 0.4 µm). The cells had one or two Golgi apparatus that stretched far into the dendrite. Mitochondria were larger here, making slender side branches that reached far into the dendrite. Due to the funnel shaped expansion, the dendrite appeared massive at its base (ø: 3-5 µm) before it leveled at a diameter of ~1 µm (ø: 0.8 – 2.0 µm, $\bar{x} \pm \sigma$: 1.1 ± 0.3 µm).

Most PLIN3s gave rise to a single dendritic branch. In five PLIN3_v a second, thin, but long branch emerged from the soma. The main dendrite was directed laterally in its initial course and descended towards the superficial layer at a shallow angle. It had an oblong cross section that was occasionally dilated in its long axis by mitochondria. The main dendrite gave rise to many side branches that projected either laterally or superficially. The branching pattern remained asymmetric and the main dendrite could be retraced over its entire length. With some distance to the soma thicker, longer side branches emerged that ramified themselves. On its course towards the superficial parts, the dendritic processes passed through large portions of the GL volume.

The proximal portions of the dendrites bore spine-like protrusions at low to moderate density. A qualitative examination by randomly visiting subsets of these excrescences suggests that these were only post- and not presynaptic. However, a considerable fraction did not receive any input. Along the proximal to distal axis of the dendrites the appearance of protrusions gradually transitioned from a more spine-like aspect to a more varicose aspect. Although no structural feature could be identified that would indicate a separation between a pre- and postsynaptic part, synaptic vesicles were not abundant in the proximal portions of the dendrites.

Presynaptic sites were most reliably found at enlargements of the distal dendrite and with considerable distance to the soma also on the shaft of the primary dendrite. The distal varicosities could arise at every part of the distal dendrite, i.e., shaft, bifurcations or at the terminal end of a branch. They were mainly bulky, amorphous with a rough surface and often bear stubby or pointy excrescences. Both pre-and postsynaptic sites were observed in these varicosities.

The Ribbon-Like PLIN3

Nine PLIN3s were categorized as ribbon-like PLIN3s. Their somata were associated with a small cluster of neurons heterogeneous in morphological types. The cell bodies had a smooth surface and sometimes made one or two spine-like protrusions of irregular shape. The nuclei had an even outline without notable indentations (Fig. 10B1). Cytosol was particularly sparse in the perikaryon of these cells. All neurons had a single dendrite with very thin and short branchlets in the proximal portion. Side branches became more substantial and ramified with increasing distance from the soma. Compared to the PLIN3_vs, the dendrites spanned a larger volume, but, because of a lower branching frequency, they filled this volume to a lesser extent. The overall appearance of this subset was more delicate, with thinner branches. The dendrite, which was oblong in cross section and predominantly smooth in surface, frequently twirled around its own axis. Some neurons of this subgroup exhibited a unique pattern of curls in their dendrite reminiscent of a ribbon (Fig. 10B6). Proximal to the soma the dendrites infrequently bore spine-like excrescences that were mainly of filopodial and short-necked, flat, round headed appearance. At the distal dendrites varicosities were far less frequent and less pronounced than in the PLIN3_vs and flattened and smooth enlargements, not bulky, rough ones were more prominent (Fig. 10B4).

The Varicose PLIN3

The somata of the ten neurons assigned to the varicose PLIN3s (PLIN3_vs) were all loosely immersed in the neuropil of the PL. Six somata exhibited a rather craggy or bumpy surface without spines. Their nuclei showed undulating indentations and cytosol was slightly more abundant. Often long-stranded ER was arranged as a single concentric circle surrounding the nucleus. Five PLIN3_v gave rise to a second, smaller neurite that were either directed towards the ventro-lateral GL or towards the PL at the dorso-lateral anterior side of the imaged volume. The dendrites had a more robust appearance: they were slightly thicker, varied in diameter and possessed an irregular, bumpy surface structure. With one exception the PLIN3_v dendrites lacked the thin, short branches observed with PLIN3_rl dendrites.

Proximally, the PLIN3 dendrites bore spine-like excrescences that were often sessile, stub–like protrusions, sometimes reminiscent of sessile filiform appendages observed in GLIN2s or they were similar in appearance to the frayed spines observed in DLIN2 (Fig. 12B). The distal portions of the dendrites were decorated with amorphous, bulky varicosities at a very high frequency (Fig. 10B5).

The dendritic morphology suggested that the PLIN3s did not restrict their neurites to a single glomerulus. The breadth of the glomerular interactions of PLIN3 was investigated by overlaying each

PLIN3 with the PNs forming a given glomerulus. The relative proportion of dendrite projected into a particular glomerulus was estimated and it was confirmed that at least one synaptic connection was made within a given glomerular volume. This examination confirmed that PLIN3s interacted with numerous glomeruli (2-8 glomeruli, $\bar{x} \pm \sigma$: 3.9 ± 1.7 glomeruli). The fraction of a dendrite innervating individual glomeruli varied between cells. Some PLIN3s projected strongly to few glomeruli, others showed a more distributed glomerular innervation pattern with varying degree of preference for individual glomeruli. No obvious difference between PLIN3_rls and PLIN3_vs with respect to the number of glomeruli or the degree to which different glomeruli were innervated could be made out.

Interactions With Other Neurons

The connectivity between PLIN3s and PNs was investigated at the example of five PLIN3_vs and two PLIN3_rls for the glomeruli GL4, GL5, GL6, GL7, GL11 and GL12. The PLIN3s were strongly connected to MCs and within most glomeruli PLIN3s connected to more than 40 % of the MCs (11/15). On average 66 % PLIN3_v-MC pairs and 83 % PLIN3_rl-MC pairs were connected (PLIN3_v: ID0893: 8/17 MC (47 %) ID1153: 30/47 MC (64 %), ID1257: 12/17 MC (71 %), ID1457: 19/40 MC (48 %), ID1504: 7/7 (100 %); PLIN3_rl: ID0952: 12/14 MC (86 %), ID3362: 11/14 MC (79 %)).
Most MC-PLIN3 pairs were reciprocally connected (PLIN3_v: 47/77 pairs, PLIN3_rl: 17/23 pairs). Four of the PLIN3_v-MC pairs were not connected by reciprocal synapses but through spatially separated in- and outbound synapses. Most PLIN3_v-MC pairs connected through reciprocal synapses made additional unidirectional synapses (37/43 pairs). These were mainly outbound synapses of the IN (30/37 pairs), but inbound synapses were also prominent (19/37 pairs). All 17 reciprocally connected PLIN3_rl-MC pairs made reciprocal synapses and in seven pairs one or two additional outbound synapses could be found. Among the unidirectionally connected pairs, PLIN3-to-MC pairs were more frequent (PLIN3_v: 26/30 pairs, PLIN3_rl: 5/6 pairs) than MC-to-PLIN3 pars (PLIN3_v: 4/30 pairs, PLIN3_rl: 1/6 pairs). The number of synapses per pair ranged from one to 11**, though** most connections involved two to four synapses. Individual PLIN3s or MCs showed no bias toward making higher or lower numbers of synapses. Furthermore, synapse number correlated neither with the PLIN3 (PLIN3_v or PLIN3_rl) or MC subtype, nor with connection directionality or the specific glomerulus innervated.
Connectivity with RCs was rather sparse for the PLIN3_vs with only one of 21 investigated pairs connected through a synapse from the PLIN3 onto the MC. In contrast, three of five PLIN3_rl-RC pairs were connected, with the PLIN3 giving input to the RC dendrite via one or two synapses. During the reconstruction process two connected PLIN3_v – RC pairs were detected. In one pair the RC provided input to a spine-like appendage of the proximal portion of the main dendrite. The other pair was reciprocally connected through three synapses (two reciprocal, one inbound to the PLIN3) at a more proximal side branch of the PLIN3 dendrite. Three more PLIN3_rl-RC pairs were observed in which the PLIN3s provided input to the RC dendrites through one or two synapses.

Five PLIN3-GLIN1 pairs were found to be connected, one reciprocally connected pair, three GLIN1-to-PLIN3 pairs and one PLIN3-to-GLIN1 pair. All connections were mediated through one or two synapses. Further, PLIN3s were found to receive input from PLIN2s and DLIN1s and provided input to GLIN2s and GLIN3s.

PLIN4

Morphology

The PLIN4s were compirsed by 33 axonless, widespread, thin branched Ins (view). The wide-spread nature of the dendritic tree, the overall thin appearance of the branches and the fact that no cell was entirely contained in the volume complicated the assessment of the neurons' completeness. Most PLIN4 cell bodies were found in PL as part of a small cell cluster or freely in the neuropil. Four somata were located in GL, one at the very surface and three others were part of larger cell clusters in GCL. The somata were ovoid or globular but usually flattened in one axis and of small to medium size (ø: 8 - 10 µm, $\bar{x}$ ± σ: 9.0 ± 0.7 µm). Their surface was uneven but did not bear any spines. The nuclei were roughly ovoid-shaped and displayed mild undulations. The perikaryon contained little to moderate amounts of cytoplasm surrounding the nucleus and was mostly devoid of cellular organelles with the exception of few small mitochondria or mitochondrial branches. The organelles collected in bulges of cytoplasm located at the base of the primary dendrites. Here, mitochondria of varying size were abundant: These mitochondria could be notably voluminous and exhibited a relatively lightly stained intermembrane space and small cisternae. Most cells contained two Golgi apparatus located at the base of the dendrite without stretching into it. In one cell the cellular organelles accumulate in bulgy enlargements of the biggest dendrite that were located at several microns from the soma. Particularly in the initial portions the dendrites were pervaded with ample, darkly stained strains of ER.

With exception to two cells, which were cut at the soma, all PLIN4s possessed a primary cilium of short to medium length (0.8 - 4.3 µm, $\bar{x}$ ± σ: 2.6 ± 0.8 µm) that arose from the soma or base of one of the main dendrites. The somata gave rise to one or two larger dendritic branches (ø: 0.3 - 1.8 µm, $\bar{x}$ ± σ: 1.0 ± 0.4 µm) and up to four smaller branches (ø: 0.1 - 0.5 µm, $\bar{x}$ ± σ: 0.4 ± 0.2 µm). The dendrites spread widely and the dendritic tree assumed any orientation on a range from fusiform to stellate. Long, thin side-branches of delicate, almost fragile appearance branched off rather infrequently. PLIN4s were distinct from other bulbar INs but they expressed high heterogeneity within the morphological class. Closer inspection revealed four morphological variants; however, these did not delineate strongly and may reflect a phenotypic continuum.

The first variant of PLIN4 comprised 9 cells the occupied mostly PL. However, most cells also projected a notable portion of their dendrite to GL. Their dendritic cross sections generally appeared round but exhibited considerable variability, often displaying an irregular circumference and surface that gave them a withered and distinctly gnarled appearance. This subgroup was also characterized by bearing many sessile protrusions that were sometimes varicose and voluminous occasionally bulbar, but more often they were flattened along one axis. The overall appearance of these enlargements was amorphous and frayed and oftentimes they made one or more filiform protrusions. The ten representatives of the second PLIN4 variant usually gave rise to a single dendritic branch that ramified predominantly in GL not PL. The somata were also positioned more superficial in PL or GL. The cross section of the dendrite was oblong and was relatively smooth. These protrusions were overall less frequent than in first subtype and while some sessile amorphous, frayed enlargements appearance were also present, protrusions resembling thorns or stomps or sometimes short-necked round-headed mushroom spines were prevalent. The third variant comprised eight neurons and was characterized by the presence of thin, delicate side-branches that usually did not bear any spine-like protrusions. On the larger branches few sessile protrusions could be observed but infrequent thorn-like or filopodial protrusions dominated. The fourth variant was dominated by a very hairy appearance that arose from the presence of many thin side-branches and the frequent elongated spine-like appendages that resembled filopodial, long-necked or branched spines. In between sessile enlargements with filopodial were also observed. The overall appearance of these appendages was frayed.

Varicosities were randomly sampled in a subset of PLIN4s starting on the proximal parts of the dendrites. Proximal to the soma varicosities were predominantly postsynaptic but frequency of outbound structures increased with distance from soma independent of the layer through which the branch projected. The transition from a postsynaptic to a presynaptic structure did not involve a markedly distinct shift in appearance of the varicosities, but by trend distal enlargements were more varicose and less frayed and bore fewer filiform protrusions or stomps.

PLIN4s were found to traverse up to five glomeruli ($\bar{x} \pm \sigma$: 2.6 ± 1.2 glomeruli). The degree of glomerular innervation was highly variable. The more superficial subtypes immersed considerable portions of their dendrite into the glomerular volumes. The deeper representatives that spread more widely through the imaged volume often immersed only a small portion of their distal branches into the volume of a given glomerulus. For three PLIN4s no association to any of the glomeruli described above was found. Two of these did not enter the GL within the imaged volume and the third reached the GL with only a single branch that passed between glomeruli.

Interactions with Other Neurons

The connectivity between PLIN4 and PNs was investigated for seven PLIN4s in the glomeruli GL1, GL5, GL6, GL7, GL11 and GL12. Contact sites with MCs were rather infrequent and not all but most contact sites had synapses. On average the PLIN4s were connected to 28 % of the MCs in the examined glomeruli (ID-998: 15/40 (38 %), ID1027: 9/33 (27 %), ID1038: 5/12 (42 %), ID1066: 8/31 (26 %), ID1199: 11/34 (32 %), ID1207: 6/20 (20 %), ID1938: 4/25 (16 %)).
Most connections involved one or two synapses (50/58 pairs), only two PLIN4s made connections with some MCs that involved three or more synapses. Reciprocally connected pairs (26/58 pairs) and PLIN4-to-MC pairs had similar prevalence (22/58 pairs). All reciprocal connections were mediated by reciprocal synapses with additional unidirectional out- (8/10 pairs) or inbound (3/10 pairs) synapses for the PLIN4 in eight pairs. In ten connected pairs the PLIN4 received input from the MC, six of these targeted the proximal portion of the PLIN4 dendrite.
During the neuron reconstruction process additional 13 reciprocally connected pairs and as well as five unidirectionally connected PLIN4-MC pairs in both directions, respectively, were identified.

PLIN4s were broadly connected to RCs within the glomeruli examined, in 15 of 36 possible PLIN4-RC pairs the PLIN4 provided input to the RC. These synapses targeted the RC dendrite and were usually quite large compared with other outbound synapses of PLIN4. In one pair the PLIN4 input was reciprocated by a synapse from the RC ruff to the more proximal portion of the PLIN4 dendrite. Most pairs were connected by only one or two synapses (11/15 pairs), in one pair, however, nine synapses onto the RC dendrite were identified.
During neuron reconstruction an additional 24 PLIN4-RC pairs were observed. These involved two PLIN4-to-RC pairs, 11 RC-to-PLIN4 pairs and 11 reciprocally connected pairs. Most reciprocally connected pairs were connected via reciprocal synapses. The number of synapses was low (one or two) in most of these pairs including all in which the PLIN4 received input but exceeded three synapses in nine PLIN4-RC pairs. Three PLIN4s made connections onto the same ruff and in close proximity an additional synapse between a pair of these PLIN4s was observed.

Furthermore, PLIN4s were found to connect to both GLIN1s and GLIN4s including both PLIN4-to-IN and IN-to-PLIN4 pairs. Output to PLIN2s and both in- and output to other PLIN4s were observed. Moreover, PLIN4s were found to give input to DLIN1s and DLIN2s and to receive input from DLIN2s.

DLIN1

Morphology

The first class of DLIN (DLIN1) comprised 61 cells (view). The key characteristic of DLIN1s was the predominance of spine-like appendages with a rounded outline of their ending. The shape of the spine-like appendages could obtain substantially different appearance: Both long and short necked spine-like appendages were present, and the spine head could appear as flattened membrane sheath or as bulky enlargement (Fig. 12A). Any graduation in between these two ends of the scale could occur. Stubby spine-like appendages sitting directly on the dendritic shaft in form of flat, enlarged membrane protrusions with rounded endings were also frequent. Thin, filopodial spines without head were regularly observed in this neuron class. Most spines showed cupped indentations, where they wrapped around a connected neurite.

The distal dendrites in GL had an oblate outline, which together with a smooth and even surface created a ribbon-like appearance and occasionally made curls (Fig. 13A4). The dendrite made frequent dilations, in which it stretched out and became so thin that in some portions the cytoplasm was no longer discernible. These dilations occurred at branch points, along the branch, and most commonly, when wrapping around connected neurites. Despite obvious similarities these processes could be easily distinguished from glial sheaths by an abundance of synaptic vesicles (Fig. 13A4, 13B2). Oftentimes, sites of synaptic interactions adopted a more complex shape: Wrapping around connected partners, they formed varicosities with cavities holding the connected neurite. Amorphous, complex structures emerged when the neuron enveloped several neurites in proximity. However, these varicosities were not only presynaptic, but also postsynaptic, and were targeted not only in reciprocal synapses, but also received unidirectional input.

Three different subclasses were distinguishable. These differed mainly in the prevailing shape and the frequency of their spine-like appendages, but also other morphological differences. These differences could be subtle and transitions between subclasses were smooth.

The Rough DLIN1s

The rough DLIN1s (DLIN1_rs) were characterized by a rough appearance of the soma and proximal dendrites, which were, often densely, covered with spine-like appendages. This subclass included the largest DLIN1 representatives and comprised 20 neurons. With exception of three cells, the medium-sized somata (ø: 8-11 µm, $\bar{x} \pm \sigma$: 9.6 ± 0.8 µm) were part of a big cluster of the GCL. Most somata were ovoid, others were roughly globular. The smooth outline of the ovoid nuclei stood in stark contrast to the rugose surface of the cell body (Fig. 13A1). The perikaryon appeared as a thin layer of cytoplasm surrounding the cell body. Only few, very small, compact mitochondria were observed. The ER was also scarce and often had an uncommonly large lumen. Depending on the number of big dendrites emanating from the soma, one or two Golgi apparatus could be found. They were in a swell at the base of the big dendrites and extended far into them. Here, a few large vesicles and lysosomes could be observed. From each soma a single primary cilium projected into the surrounding extracellular space, usually directed parallel to the cell body's outline (length: 2.0-3.3 µm, $\bar{x} \pm \sigma$: 2.9 ± 0.6 µm).

The surface of the somata was conspicuously rough and craggy: It was overcast by flat, extensive membrane excrescences and protuberances, and covered by many rounded spine-like appendages

(Fig. 13A1,2). In addition, somata often showed indentations from passing neurites. They received input that primarily targeted the bulgy excrescences, but also the soma directly. Further, protrusions of neighboring cell bodies regularly penetrated the somata and made extended cavities. Only a subset of these structures was associated with a synapse. DLIN1_rs were not observed to make such projections into another cell but some somata made synapses onto passing neurites.

Usually, DLIN1_rs had one or two large dendrites, directed towards the GL, and one or two additional small, basal dendrites that remained in the deep layers. Cells with a single large dendrite (12/20) or several dendrites of equal size (8/20) emerged one or more small, basal dendrites from their large, primary branch. The appearance of large and small dendrites differed significantly. Due to copious membrane excrescences on their surface, the large primary dendrites had a similarly rough and craggy appearance as the soma. This aspect was further intensified by the dense coverage with predominantly short-necked and stubbed spine-like appendages, which could be observed over several tens of microns on the proximal parts of the dendrite (Fig. 13A2). The irregular cross section complicated precise measurements of the dendritic diameter, which varied between 1 – 4 µm ($\bar{x} \pm \sigma$: 2.2 ± 0.7 µm). In all DLIN1_rs at least one large branch was directed towards the GL at the ventro-lateral side. Projections towards the dorso-lateral and dorso-medial side were also common, presumably targeting GL areas in other parts of the OB. In DLIN1_rs with a single main dendrite at least one symmetrical branching occurred. Noteworthy is the extent of the dendrite, which spanned the entire imaged volume in its long axis, whilst still being notably cut in most DLIN1_rs.
The frequency of spine-like appendages gradually reduced on the trajectory of the main dendrites towards the GL, and upon entry in the GL they were absent. Traversing the deep layers, the large dendrites made several side branches with the same morphology as the deep, basal branches. This asymmetric branching scheme was no longer dominant within GL. Here, the dendrites branched heavily, spread widely and extensively, and displayed distal varicosities as described above (Fig. 13A). These distal enlargements were the primary sites of reciprocal synapses with MCs. Consistent with sides of strong synaptic interactions, mitochondria were abundant here. The cytosol of the distal dendrites was filled with vesicles. Synapses could be easily distinguished by a cloud of small vesicles and membrane fusion sites. In addition, the dilated enlargements frequently contained large vesicles with clear lumen. These were abundant in the distal dendrites, but clear signs of active zones site or vesicle fusion sites were lacking.

The basal dendrites were very thin, with a diameter of 0.3 – 0.75 µm ($\bar{x} \pm \sigma$: 0.5 ± 0.2 µm). As the apical processes, they were oblong in cross section. They usually projected dorso-laterally and dorso-medially within the deeper layers. Though their length could exceed 50 µm, they did not enter GL. The basal dendrites also bore spine-like appendages, but at a much lower frequency (Fig. 13A3). This frequency decreased further with distance to soma. The spine-like appendages were usually long-necked or thin, rarely stubby. It was not uncommon to find spine heads receiving more than one input. An inspection of multiple, randomly picked spine-like appendages on the proximal large dendrites and the basal, deep layer dendrites showed that spine heads were mostly devoid of vesicles and no presynaptic structures were observed. ER was frequently found in spine heads, however, these structures did not exhibit the stacked structure reported for mammalian spine apparatus.

Before dendrites entered the GL, mitochondria were sparse and rather small in the thinner branches. Microtubuli were clearly visible in the cytosol throughout the dendrite, even in thinner ones. The cytosol often contained larger vacuoles, and very dark, threadlike ER was abundant. Few vesicles were observed in the basal and proximal portions of the large dendrite. A few presynaptic sites were found on the dendrite shafts. The small dendrites in the deep layer received a lot of synaptic inputs. In the basal dendrites and the proximal portions of the main dendrites, this input was mostly directed to the spine-like appendages. With increasing distance to the soma the main dendrites received more input to the shaft.

The DLIN1s with Long-Necked Spines

The predominance of long-necked spine-like appendages characterized the second subclass of DLIN1s (DLIN1_l), a group comprising 22 neurons. On a coarse morphological scale DLIN1_ls differed from DLIN1_rs because they were smaller, and both the location of their soma and the extent of their neurite tree were more superficial.

Most DLIN1_ls somata located in the PL, either as part of a small cluster or freely in the neuropil whereas five cells were part of a large cell cluster in GCL. The somata were slightly smaller than DLIN1_rs in terms of their largest diameter (ø: 8-10 µm, $\bar{x}$ ± σ: 9.0 ± 0.7 µm), but more so in volume. An irregular, roughly globular soma outline was predominant, and all nuclei displayed mild undulations. The soma surface was slightly uneven and bumpy, but spine-like appendages were infrequent and membranous excrescences observed in DLIN1_rs were absent. A single primary cilium of medium length was found in all DLIN1_ls (1.5 - 3.3 µm, $\bar{x}$ ± σ: 2.4 ± 0.5 µm).

The soma gave rise to one to four dendrites projecting to GL ($\bar{x}$ ± σ: 1.6 ± 0.8). The separation between small, basal and large, distally projecting neurites was far less pronounced than in DLIN1_rs. Three cells completely lacked basal dendrites completely; in other three more they resembled a branched version of the long-necked spine-like appendages. Most DLIN1_ls, however, gave rise to a few basal dendrites of variable length (≤ 15 µm or > 50 µm, ø: 0.3 - 0.8 µm, $\bar{x}$ ± σ: 0.4 ± 0.1 µm), which projected through the deeper layers (Fig. 13B1). They made thin and long-necked spines at long, but regular intervals. Synaptic vesicles were absent from the basal dendrites as well, but many postsynaptic sites at which input from thin neurites of remote cells seemed to dominate. The distally directed dendrites emerged with relatively small diameter (ø: 0.5 – 2.0 µm, $\bar{x}$ ± σ: 1.1 ± 0.4 µm). Neurons with a soma position deep in PL or GCL tended to expand their dendritic trees more radially, spreading widely, before projecting towards the ventro-lateral GL. The dendritic orientation of the other cells followed the long axis of the imaged volume. Their ramification in the GL was strongest and represented the larger fraction of their dendritic tree.

The larger dendrites sometimes bore stubby and short-necked spine-like appendages in their proximal parts. The predominant form of appendages was, however, the long-necked variant with a flat or a bulky rounded head. These covered the dendrites at moderate frequency, which only slightly decreased towards the GL. The apical dendrite was characterized by sheath-like expansions that could be platelet-like, cupped or obtain more complex shapes enveloping other neurites. The distal varicosities occurred at a higher frequency than in the DLIN1_rs. They were by trend smaller in diameter and more circular. A noteworthy fraction took on a spine-like appearance, resting on neck-like protrusions. The presence of synaptic vesicles distinguished them from deep layer spine-like appendages.

The Sparsely Spiny DLIN1s

The sparsely spiny DLIN1s (DLIN1_s') comprised a group of 19 thin-branched DLINs. Their cell bodies were in big clusters of the GCL (10/19), in PL (9/19) as part of small clusters or immersed in the PL neuropil. Somata were of medium size (ø: 8 - 12 µm, $\bar{x}$ ± σ: 9.8 ± 0.9 µm) and predominantly of irregular, roughly globular shape. The surface was coarse, bore isolated short-necked spine-like appendages and membrane excrescences. These were, however, far smaller, and less pronounced than in the DLIN1_rs. As in the other DLIN1 subclasses, primary cilia were of medium length

(1.8 - 4.0 µm, $\bar{x} \pm \sigma$: 2.5 ± 0.6 µm). The somata gave rise to two to four slender dendrites (ø: 0.3 – 2.0 µm, $\bar{x} \pm \sigma$: 1.2 ± 0.4 µm). Only four neurons showed a clear size difference of dendrites emerging form the soma. Small, thin dendrites, which restricted their projection to the deep layers, could be found in other DLIN1_s' as well. But, apart from the small diameter, differences to the other dendrites were not obvious. The dendrites were sparsely studded with stubby and short-necked spine-like appendages dominating. Only in three DLIN1_s' long-necked variant seemed to prevalent. As in the other subclasses, the dendrite branched more frequently in the GL and displayed varicosities of medium frequency.

The overlap of the dendritic tree with identified glomeruli in the imaged volume was inspected for all DLIN1s, but the presence of synapses within the glomerular volume was only confirmed for a subset. All cells sending a dendrite into the GL within the imaged volume projected through the volume of at least one glomerulus. The fraction of distal dendrites immersed in individual glomeruli was highly variable within and across neurons. Representatives of different subclasses projected to a similar number of glomeruli (DLIN1_r: 1-7 glomeruli, $\bar{x} \pm \sigma$: 3.5 ± 1.6, DLIN1_l: 1-5 glomeruli, $\bar{x} \pm \sigma$: 3.0 ± 1.5, DLIN1_s: 1-6 glomeruli, $\bar{x} \pm \sigma$: 3.1 ± 1.3 glomeruli).

Interactions with Other Neurons

The interactions with PNs within a glomerulus were investigated for two representatives of the DLIN1_rs, DLIN1_s' and DLIN1_ls, respectively, in the glomeruli GL6, GL9, GL11 and GL12. Only one DLIN1_r projected to all four glomeruli.
This analysis suggested a complex and individual pattern of interaction between MCs and DLIN1s. All examined DLIN1s made synapses with at least one MC within each of the glomerular subcompartments to which they projected. The fraction of MCs connected to a given DLIN1 varied per glomerulus, and a given DLIN1 could be connected to >40 % of the MCs in one glomerulus and contact MCs more selectively in another. With four in six and three in five selectively innervated glomeruli, respectively, the DLIN1_rs and DLIN1_s' were more selective than the DLIN1_ls (1/5). This was also reflected in a higher overall connection rate between DLIN1_ls and MCs (DLIN1_r: ID1602: 7/27 MCs (26 %), ID3212: 18/58 MCs (31 %), DLIN1_l: ID0956: 15/34 MCs (44 %), ID1521: 9/19 MCs (47 %), DLIN1_s: ID0246: 8/19 MCs (42 %), ID3252: 10/46 MCs (22 %)). Reciprocal connections were predominant in all subclasses, however, the DLIN_s' made a similar number of outbound connections with MCs (DLIN1_r: reciprocal: 15/25 (60 %), outbound: 5/25 (20 %), inbound: 5/25 (20 %), DLIN1_l: reciprocal: 17/24 (71 %), outbound: 4/24 (17%), inbound: 3/24 (13 %), DLIN1_s: reciprocal: 9/18 (50 %), outbound: 8/18 (44 %), inbound: 1/18 (6 %)). The number of synapses was low: Only 15 % of the DLIN1-MC pairs was connected by three or more synapses. With two exceptions making spatially separated in and outbound synapses, reciprocal connections were mediated by reciprocal synapses with some pairs making additional in or outbound synapses (14/39 reciprocally connected pairs).
A noteworthy aspect of the DLIN1-MC connectivity was the occurrence of very large synapses from a DLIN1 onto an MC. These involved a pronounced dilation of the distal DLIN1 dendrite forming a flattened sheath around one or more MC dendrites. These DLIN1 varicosities were filled with dendrites and the could comprise several active zone or the site of vesicle fusion could extend over several 100 nm2. These synapses were only observed in reciprocal contacts between DLIN1_r and MCs (Fig. 14B).

Representatives of all three subclasses were found to connect to RCs. The frequency of connections varied notably between the three subclasses. While the DLIN1_rs and the DLIN1_ls connected broadly to the majority RCs within the glomeruli examined only one of the DLIN1_s-RC pairs was connected (DLIN1_r: 7/12 (58%), DLIN1_l: 7/10 (70%); DLIN1_s: 1/8 (13 %)). In most connections (14/15) the DLIN1s provided input to the dendrites of the RCs through one to four synapses. One DLIN1_r received input from an RC. Observations of connections during neuron reconstruction extend the picture of DLIN1-RC connectivity by an additional 37 connected pairs involving all DLIN1 subclasses. Outbound connections onto the RC dendrite dominated but in particular the DLIN1_s made many connections at the RC ruff which involved predominantly reciprocal connections but also some inbound connections.

All synapses to PNs in the GL originated from the varicose enlargements on distal dendrites. Both the proximal portions of the apical dendrite and basal dendrites received many synapses onto shaft and spines. The somata received input from neighboring cell bodies and from thin dendrites with dark staining full of small, granular vesicles. Most of the input terminating on the proximal parts of the DLIN1s originated from neurites coming from deep sources. During proofreading diverse interactions of DLIN1s with the local IN network were observed.

Connections between DLIN1s, as well as to representatives of other DLINs, were found not only in the deep layers, but also in GL. These synapses were located on the varicosities of distal dendrites, often in vicinity to MC dendrites, to which both DLINs connected. These connections involved unidirectional in- or outputs but also a reciprocal synapse between two DLIN1_rs was observed. These DLIN pairs were connected through a single synapse. DLIN1s were found to make outbound and reciprocal connections with GLIN1s. Both DLIN1_rs and DLIN1_s' were found to receive GLIN2 input, and several outbound connections form DLIN1_ls onto GLIN2 were detected. Outbound synapses onto GLIN3s, GLIN4s and PLIN3s were observed and input from PLIN4s.

DLIN2

The 81 DLIN2s were characterized by the presence of spine-like appendages of a unique bulgy and frayed appearance with high variability in the overall shape (Fig. 12B, view). Short necked appendages were the dominating type. Elongated, long-necked spine-like appendages reaching several microns in length were occasionally observed, sometimes featuring side branches with pointed tips. Because these elongated protrusions were morphologically difficult to distinguish from small dendritic branches, any studded, deep-layer protrusion under a visually judged length of five microns was classified collectively as a spine-like appendage. The necks could be very narrow (≤ 100 nm), which contributed to the overall tattered appearance of the spine-like appendages. In their simplest form, spine heads resembled bulky rods or bulbar enlargements. In most cases, they were completely amorphous and studded with several protrusions. These protrusions could be short pointy tips, more filopodial, or of beaded appearance, due to repeated bulbar dilations followed by constrictions. Stubby spine-like appendages were the second most common type. They often sat on dendritic varicosities, had a large base, and bore many short, filopodial protrusions. The heads of the appendages seemed to be mainly postsynaptic to a single contact. Rarely, multiple synapses onto the same spine-like appendage were observed. In addition, a small, but notable fraction did not receive any synaptic input. Synapses onto the filopodial protrusions were not observed. The heads often contained vesicles and ER strands, that did however, not form layers of stacked discs as commonly described for spine apparatus.

The distal dendrites were very thin (ø: ~ 150 - 500 nm). Pre- and postsynaptic interaction mainly occurred at bulky or flattened varicosities, which could assume a complex, amorphous shape, when coiling around connected neurites. Some enlargements were similar to those observed in DLIN1s, they were flattened and stretched to the extent that they appeared perforated. More often, the varicosities were bulky, but elongated along the branch, resulting in a roughly spindle-shaped structure, sometimes indented by connected neurites. Furthermore, the varicosities frequently bore filopodial, pointy protrusions, extending into the neuropil without synapses. The cross section of the dendrites had a roundish outline, which varied a lot in diameter, giving rise to a rather bumpy, irregular surface. Dendrites showed morphological features of distal dendrites already in the PL. Both small synaptic vesicles, as well as larger vesicles with clear lumen, were abundant in the varicosities and branches of the distal dendrites.

Morphology

The Common DLIN2s

The largest of the three morphological subclasses of DLIN2s, the DLIN2_cs, comprised 52 neurons whose somata were predominantly located in big clusters of the GCL (44/52). Seven neurons had their cell bodies located in a small cluster or freely in the neuropil of the PL. The overall size, i.e., soma size, neurite perimeter and length gradually reduced from deep to more superficially located cells. The somata were among the largest observed for INs of the OB (ø: 9 - 11 µm, $\bar{x} \pm \sigma$: 10.0 ± 0.8 µm) and were mainly pyramidal to globular-shaped or rather ovoid. Their surface was mostly even and studded with isolated spine-like appendages or membrane excrescences. The somata usually exhibited impressions of neighboring somata. The nuclei were embedded in moderate amounts of cytoplasm and displayed only minor undulations (Fig 15A1). Their perikaryon contained little to moderate amounts of long stranded ER. The mitochondria were slender and often elongated. Many multi-vesicular bodies and darkly stained endosomes were observed in vicinity to the Golgi apparatus. These varied strongly in size; they were mostly located at the base of primary dendrites and sometimes stretched into them. Primary cilia of considerable length (2–8 µm, $\bar{x} \pm \sigma$: 3.9 ± 1.1 µm) were found in all neurons (Fig 15A2, black arrow). They projected into the surrounding extracellular space without any obvious common direction.

Two to four dendrites of small to medium diameter emerged from the soma. About half of the cells (25/52) exhibited a distinct size difference in their primary dendrites ($ø_{LARGE}$: 0.8 - 3 µm, $\bar{x} \pm \sigma$: 1.6 ± 0.6 µm, $ø_{THIN}$: 0.2 - 0.8 µm, $\bar{x} \pm \sigma$: 0.6 ± 0.2 µm, equal-sized: ø: 0.5 – 2.0 µm, $\bar{x} \pm \sigma$: 1.1 ± 0.4 µm), but apart from diameter the wider and narrower branches did not differ morphologically. The dendrites branched regularly, making both symmetric and asymmetric bifurcations. This resulted in a predominantly stellate orientation, with branches spreading widely through the deep and superficial portions of the imaged volume. In contrast to DLIN1s, the branching frequency increased already within the PL. The dendritic cross section was round. Thin, elongated mitochondria and ER strands with wide lumen stretched through the proximal dendrites in GCL. Disregarding the spine-like appendages, the surface structure of the proximal branches was mainly even and smooth. At more or less regular intervals the dendrites of the deep layers expanded in diameter and formed varicosities, often, but not always, associated with an accumulation of mitochondria (Fig. 15A1). In most cases, these enlargements formed a regular expansion of the dendrite to all sites, but they could also obtain more irregular shapes e.g., stubby or cone-like. On the proximal dendrites a large fraction of the varicosities was associated with spine-like protrusions (Fig. 15A2, A4). These varicosities were not associated with synaptic vesicles. In the more superficially positioned DLIN2_cs, varicosities on proximal dendrites were less pronounced or absent. Several outbound synapses from the dendritic shaft could be observed on the proximal dendrites.

Completely aspiny dendrites with a conspicuous beaded appearance were occasionally present. Instead of typical varicosities, these branches featured highly frequent, small roundish enlargements (0.5–1.0 µm). Interconnected by thin, short dendritic segments, they gave the processes a distinctive pearl-necklace-like morphology (Fig. 15A3). The varicosities did not contain mitochondria, but the dendrites were rich in ER. In contrast to what was common in thinner neurites, microtubules were very clearly visible in these dendrites. At a low frequency, both pre- and post-synapses were found at the varicosities. These beaded dendrites were guided through neuropil-rich regions of the GCL or the PL.

Spine-like appendages were not restricted to dendritic varicosities and could emerge directly from the shaft as well. The frequency of spine-like appendages showed an almost bimodal distribution: About half of the DLIN2_cs were only sparsely spiny, almost aspiny on the main dendrites, and made only short, studded appendages. The other DLIN2_cs displayed a moderate to high frequency of spine-like appendages that were predominantly stubby or short-necked, with only a few representatives showing some long-necked appendages.

The transition from proximal, deep dendrite to distal dendrite was gradual. With distance from the soma frequency of spine-like appendages decreased, while many times the frequency of the varicosities increased. They became smaller in diameter and were no longer associated with spine-like appendages. At these varicosities, in- and outbound, as well as reciprocal, synapses were found already in PL (Fig. 15A5). Upon entrance into the GL varicosities became larger, bore pointy protrusions, and more often adopted a flattened appearance as described above.

The Fringy DLIN2s

The DLIN2s with fringy spines (DLIN2_fs) comprised 22 neurons. The globular somata (ø: 9 – 12 µm, $\bar{x} \pm \sigma$: 9.9 ± 0.7 µm) were part of a big cluster or immersed in the neuropil-rich regions of the GCL. Primary cilia were long (2.3 – 6.0 µm, $\bar{x} \pm \sigma$: 3.6 ± 1.1 µm). Up to four dendrites of small to medium size (equal-size (12/21): ø: 0.5 – 3.0 µm, $\bar{x} \pm \sigma$: 1.2 ± 0.6 µm; unequal sized (9/21): $ø_{LARGE}$: 0.8 – 3.0 µm, $\bar{x} \pm \sigma$: 1.5 ± 0.7 µm, $ø_{THIN}$: 0.3 - 0.8 µm, $\bar{x} \pm \sigma$: 0.5 ± 0.2 µm, one cell was cut at the soma) emanated from the soma, branched regularly and spread in all directions. The somata had a very rugose surface: They were studded with filopodial protrusions and many long-necked frayed spines producing a very rough, frazzled, sometimes spiky appearance (Fig. 15C2). This aspect extended to the proximal part of the dendrites. At moderate to very high density, dendrites were covered with long-necked frayed spines (Fig15C2, C3). Many spines terminated in an elongated, frayed spine head, made side branches, or bore long filopodial protrusions. Such protrusions were also found on the rarer, short-necked spines. The bulbar varicosities on the proximal dendrites associated with spines, observed in DLIN2_cs, were either absent or far less pronounced (Fig. 15C3). However, the beaded side branches, traversing the PL and neuropil rich parts of GCL, were also present in this subclass (Fig. 15C4). The appearance of the distal dendrites was indistinguishable from those of DLIN2_cs.

The DLIN2s with Stubby Spines

The DLIN2s with stubby spines (DLIN2_s') were comprised of seven neurons. All cell bodies were part of a big cluster in GCL with a diameter of 10 µm (10 – 11 µm, $\bar{x} \pm \sigma$: 10.1 ± 0.2 µm). The surface of soma and proximal dendrites was rough, rugged, and covered with stubby, frayed spines and large, elongated membrane excrescences (Fig. 15B1). DLIN2_s bore a single primary cilium of medium length (2.5 – 4 µm, $\bar{x} \pm \sigma$: 3.6 ± 0.6 µm). One or two large dendrites (ø: 1 – 3 µm, $\bar{x} \pm \sigma$: 1.8 ± 0.6 µm). arose from the soma and were directed towards the superficial OB layers on the dorso-lateral and ventro-lateral portions of the imaged volume. Additional one or two smaller, dendrites were present in four DLIN2_s' (ø: 0.5 – 1 µm, $\bar{x} \pm \sigma$: 0.7 ± 0.2 µm). The proximal dendrites were densely covered with stubbed and short-necked frayed spines but did not exhibit any varicosities (Fig. 15B1). Four cells gave rise to thin basal dendrites emerged, which bore thin or long-necked spines at low frequency or were completely aspiny. The thin dendrites ramified in the deep layers, one reaching the GL. Beaded side branches, projecting through the deeper layers were observed in four DLIN2_s. The distal dendrite showed flattened, stretched varicosities at moderate frequency (Fig. 15B3).

The overlap with the glomeruli was checked for all neurons of this class, the presence of synapses within these volumes was confirmed for a subset of DLIN2s (see below). Almost all cells with projections in the GL immersed their neurites in at least one glomerulus (DLIN2_c: 1-10 glomeruli, $\bar{x} \pm \sigma$: 3.3 ± 1.9, DLIN2_f: 2-6 glomeruli, $\bar{x} \pm \sigma$: 3.9 ± 1.1, DLIN2_s': 1-11 glomeruli, $\bar{x} \pm \sigma$: 5.5 ± 3.2 glomeruli). Because of their size and widespread dendrites, DLIN2s were often incompletely represented in the data volume. Considering only cells with CD1 the number of innervated glomeruli was significantly higher: DLIN2_c: 2-10 glomeruli, $\bar{x} \pm \sigma$: 5.4 ± 2.4, DLIN2_f: 4-6 glomeruli, $\bar{x} \pm \sigma$: 4.7 ± 0.8, DLIN2_s': 6-11 glomeruli, $\bar{x} \pm \sigma$: 8.5 ± 3.5 glomeruli.

Interactions With Other Neurons

The connectivity between DLIN2s and PNs was examined in the glomeruli GL1, GL3, GL4, GL5, GL6, GL7, GL8, GL9, GL11 and GL12 for two DLIN2_cs and DLIN2_fs, respectively, and one DLIN2_s. For four of five DLIN2s this set of glomeruli comprised all glomeruli innervated in the imaged volume. Only one DLIN2_c had projections into all ten glomeruli (example of connections with glomeruli is shown in Fig. 16). The size of the dendrite projected into a given glomerulus was highly variable and often involved only a relatively short branch. In consequence, selective innervation of glomeruli prevailed in all DLIN2 subclasses (10/12 glomeruli- DLIN2_c pairs, 4/7 glomeruli- DLIN2_f pairs, 3/5 glomeruli- DLIN2_s pairs). Nevertheless, all DLIN2s examined innervated at least one glomerulus broadly in which they connected to > 40 % of MCs. On average the connection rate to MCs was low in all subclasses (DLIN2_c: ID2319: 19/96 MCs (20 %), ID2334: 14/54 MCs (26 %), DLIN2_f: ID0222: 9/33 MCs (27 %), ID1188: 9/35 MCs (26 %), DLIN2_s: ID2366: 19/65 MCs (29 %)). For DLIN2_cs and DLIN2_fs most connections were reciprocal (DLIN2_c: 25/33 (76 %), DLIN2_f: 13/17 (76 %)), mediated by reciprocal synapses and fewer connections were outbound (DLIN2_f: 6/33 (18 %) , DLIN2_f: 3/17 (18 %)) and in very few connections only the MC provided input to the DLIN2 (DLIN2_c: 2/33 (6 %), DLIN2_f: 1/17 (6 %)). In DLIN2_s' outbound connections (10/17 (59 %)) dominated over reciprocal ones (7/17 (41 %)) and no uniquely inbound connection was found. The synapses number per connected pair was low for all DLIN2 subclasses and only eight of 69 pairs were connected by three or more synapses, here, however, up to eight synapses per pair could be found.

The DLIN2_cs connected to nine of 33 RCs (27 %). Two of these connections (20 %) were reciprocal with one involving a reciprocal synapse and the other through spatially separated in and outputs. In six connections the DLIN2 provided input to the RC dendrite (60 %) and two connections were inbound (20 %). Two of 12 (17 %) DLIN2_s-RC pairs were connected and, in both connections, the DLIN2_s provided input to the RC dendrite through a single synapse. Among the glomeruli-DLIN2 pairings examined none of the 14 DLIN2_f-RC pairs were connected.

Additional connections between representatives from all subclasses of DLIN2s and RCs were observed, for DLIN2_cs and DLIN2_fs these involved connections of all signs, whereas for DLIN2_s only outbound connections were observed.

Further several connections of DLIN2s with local INs were observed: DLIN2s made outbound synapses from shafts of the proximal dendrites and soma onto DLINs of all classes. Interactions with other DLINs were not restricted to the deep layers - several instances of synapses from DLIN1s giving input from the distal dendrites of DLIN2s were observed as well as a reciprocal synapse with a DLIN3. These synapses were in proximity to synapses with PNs to which both DLINs connected. Furthermore, instances of both in- and outbound connections with all classes of GLIN1s, GLIN3s GLIN4s and PLIN4s were found, as well as outbound connections with GLIN2s.

DLIN3

Morphology

The third class of DLINs comprised a group of 38 neurons (view). These cells stood out by their hairy appearance: All neurites in the deeper layers were densely covered with thin, elongated protrusions of several microns' length. The appearance of spines was quite diverse (Fig. 12C). Most spines were long-necked, branched, or both. Filopodial, as well as mushroom-like branches could arise in the middle of a spine neck, as well as the distal end of a spine. Many spines also exhibited amorphous dilations along the spine neck, which were usually targeted by synapses, and continued with a filopodium. Given this appearance, spines were often undistinguishable from side branches of basal dendrites, which were also very thin and delicate. The spine heads expressed a considerable variability; bulky heads dominated over flattened ones. They could be amorphous and associated with additional filopodia, pointy tips or beaded branches that tapered off. The presence of long beaded protrusions on spine heads separated the DLIN3s into a group with a frizzly appearance and a non-frizzly variant, in which these kinds of protrusions were absent, and spines were more compact. Spine heads resembling long, amorphous rods were found on thinner spine necks of different length were common. Long, filopodial headless spines were also common while stubby spines were rare.
The distal dendrites were similar to those of DLIN2s: The neurites were round in cross section and with frequent, irregular modulations of the diameter. Compact and bulky varicosities associated with pointy tips dominated. In contrast to DLIN2s, varicosities stuck out more from a branch, resembling sessile spines and often sat on the end of a short branch. Stretched out flattened enlargements were rarer.

Most cell bodies (28/38) were in the big clusters of the GCL with the remaining cells located in the neuropil of the PL. Soma size and dendrite perimeter decreased as a function with soma position along the deep – superficial axis. The somata were globular or droplet-shaped and of a medium diameter (ø: 8 – 10 µm, $\bar{x} \pm \sigma$: 8.9 ± 0.3 µm, Fig. 17A). They stood out for their mostly very smooth surface, which was yet studded with many elongated spines and delicate basal dendrites. The nuclei displayed mild undulations and were embedded in little, sometimes moderate amount of cytoplasm. The cytosol was very clear and lightly stained, containing only a few short strands of ER with wide lumen, few slender mitochondria and scarcely any vesicles. A single compact Golgi apparatus was usually located at the base of one of the larger dendrites without stretching into it (Fig. 17A2). Here, other components of the endomembrane system were identified. All neurons had one primary cilium of medium length (2 – 5 µm, $\bar{x} \pm \sigma$: 3.4 ± 1 µm).
DLIN3s gave rise to one or two primary dendrites (ø: 0.5 – 2.0 µm, $\bar{x}$: 1.1 ± 0.4 µm) and in addition most cells made up to three fine, basal dendrites (ø: 0.1 - 0.5 µm, $\bar{x}$: 0.4 ± 0.2 µm). The primary dendrites were directed laterally and superficially towards the GL. On this path, some branched once

or twice symmetrically within PL, giving rise to both either more stellate or more fusiform orientation of dendritic trees. DLIN3s projected to fewer glomeruli than other DLINs (1 – 5, $\bar{x} \pm \sigma$: 2.2 ± 1.4 glomeruli, considering only CD1 cells: 2-5, $\bar{x} \pm \sigma$: 3.4 ± 1.1 glomeruli).
Spines covered the dendrites at a very high frequency, their occurrence gradually decreased with distance from soma. In between spines the dendrites generally exhibited an even, smooth surface and were round in cross section. In 16 the proximal portion of the primary dendrites exhibited bulbar dilations at low frequency which often contained mitochondria. In a subset of 11 DLIN3s aspiny, beaded neurites (see DLIN2) traversing PL were observed. Within the deep layers, the large, superficially projecting dendrites gave rise to many delicate basal dendrites. These were densely covered with elongated spines and extended usually around 10 – 30 µm but may occasionally spread well beyond 50 µm.

In eight cells 10 – 15 spines on deep layer dendrites were randomly selected and inspected for synapses. The majority of spines received input at least at one site. Branched spines often received more than one synapse. Most spine heads were free of vesicles, but contained ER, which was not arranged in stacked discs as reported for spine apparatus. In four DLIN3s outbound synapses onto deep layer neurites in one to three spines were observed (Fig. 17A3-5). These spines were quite proximal to the soma, showed no obvious differences to neighboring spines and could be postsynaptic to more than one neurites. Further spines containing many small vesicles, which resembled synaptic vesicles were observed in which, however no sufficiently distinct sites of vesicle fusion were found. Few outbound synapses on dendrites and varicosities in the deep layers were observed.

Interactions With Other Neurons

The connectivity between DLIN3 was investigated for six DLIN3 in the glomeruli GL9, GL7, GL11 and GL12. The DLIN3s investigated connected selectively to MCs in the glomerulus or glomerular subcompartment they projected to. On average 41 of 144 possible pairs were connected, 25 of these connections were made by two DLIN3s with pronounced ramifications in one glomerulus each (ID0111: 1/17 MCs (6%), ID979: 15/23 MCs (65%), ID1084: 7/44 MCs (16%), ID1496: 10/23 MCs (43%), ID2519: 6/29 MCs (21%), ID2711: 2/8 MCs (25%)). Most connected pairs were reciprocally connected (30/41). Reciprocity was usually mediated through reciprocal synapses (28/30) of which a subset of DLIN3s made additional outbound synapses to the MC (9/28). In six pairs the DLIN3 received input from the MC through additional synapses. Many pairs were connected through a single reciprocal synapse (12/28), but up to 11 synapses per pair were observed ($\bar{x} \pm \sigma$: 3.3 ± 3.0 synapses/pair, Fig, 17B2). There was no evidence of individual DLIN3s or MCs displaying a bias toward higher or lower synapse numbers. In seven connected DLIN3-MC pairs the IN gave input to the MC and four pairs in which the DLIN3 received MC input were found. These connections were mediated by one or two synapses. Synapses onto MCs were notably large, mainly with respect to the associated vesicle cloud, but sometimes also with respect to the size of the active zone

Three DLIN3s connected to RCs in the glomeruli examined (ID0111: 1/1 RCs (100%), ID979: 1/2 RCs (50%), ID1084: 1/5 RCs (20%), ID1496: 0/3 RCs (0%), ID2519: 0/4 RCs, ID2711: 0/1 RCs). Synapses with RCs involved both reciprocal synapses at the ruff (fig.7 1B3), as well as outbound synapses onto the RC dendrite (Fig. 17B1). These could co-occur within a given connected RC-DLIN3 pair. Few additional synapses to RCs were found during the reconstruction process.
Some partners receiving input from proximal spines of DLIN3s were identified as DLIN1_s' and neurites belonging to other INs of the deeper layers e.g., PLIN4s. During reconstruction few

connections to DLINs of the other classes, as well as some connections to GLIN1s and GLIN3s were noted.

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